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Numerical Study of Cavitating Flow Over a Hydrofoil

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Numerical Study of Cavitating Flow Over a Hydrofoil

présenté par **Mohammad VASFI MARANDI**

en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*

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DEDICATION

To the memory of my father, Mehdi Vasfi Marandi (1963-2020),

and my mother Manijeh Nilfroush,

who gave me their support throughout my life and paved my way to success.

ACKNOWLEDGEMENTS

First and foremost, I would like to sincerely thank my research director, François Guibault, for his availability and guiding me during my research, and all of his unconditional supports during the tough and sad COVID-19 pandemic to complete this master successfully. I also wholeheartedly appreciate my other supervisor, Jean-Yves Trépanier, who also helped and guided me patiently during my program and supported me a lot throughout this journey. They inspired me to do my best in everything including my research project. Of course, the time was limited for learning from these two professors who are oceans of knowledge and experience.

I really appreciate Bernd Nennemann and Melissa Fortin for taking the time to participate in several group meetings and guiding me from the first steps of this research project. Their industrial point of view as well as continuous guidance of my supervisors were very useful to start my project in the best way and helped me a lot to keep working efficiently.

I am very grateful for being supported by my friend, Amir Modarreszadeh, who was a PhD student in McGill university at the time. I sincerely appreciate him for the time and energy he spent to help me elaborate knowledge in the CFD field.

Many thanks for the frequent technical supports of the Génie Informatique et Génie Logiciel (GIGL) team, especially Jonathan Brien and Jean-Marc Chevalier. They are expert and knowledgeable in their career, and they were always there whenever I was faced with any software or hardware computer problem.

I want to also thank all the employee and management team of the CEPSUM for making it possible for every student in the Université de Montréal to experience a very high quality and memorable sport activities besides benefiting from one of the most high-quality scientific services.

Last but not the least, I would like to thank my parents, my sister, all my friends, and my colleagues in the Analysis, Design and Optimization of Turbine (ADOPT) laboratory in the École Polytechnique de Montréal and everybody from near or far who accompanied me and made this journey so memorable. Absolutely, without their company, I could not have completed this program as good as the current way.

RÉSUMÉ

En raison de la complexité du champ d'écoulement cavitant et des fluctuations soudaines de pression autour des aubes des machines hydrauliques, la modélisation de la cavitation est un problème très difficile, et comme elle a de nombreux effets néfastes dans de nombreuses industries telles que l'industrie hydraulique, cela a été un sujet de recherche très actif au cours des dernières décennies. Étant donné que les variations de turbulence près d'un profil hydraulique peuvent entraîner une sous-estimation dans la prédiction de la pression locale, des pressions inférieures à la pression de saturation peuvent être prédites. Par conséquent, le choix du bon modèle de turbulence qui modélise la viscosité de manière appropriée est d'une grande importance. La recherche actuelle est effectuée à l'aide du solveur commercial ANSYS-Fluent 16.1 et elle est consacrée à la modélisation des écoulements cavitants à différents nombres de Thoma (σ) autour d'un profil NACA66. À cet égard, dans le chapitre 3, le modèle de cavitation Zwart-Gerber-Belamri couplé à l'approche de modélisation multiphase mixte ainsi que le modèle de turbulence standard $k - \varepsilon$ (SKE) est appliqué, et la précision des prédictions numériques effectuées est étudiée. Ensuite, l'effet de certains autres paramètres sur la simulation en régime permanent tels que la densité du maillage, les critères de convergence, divers modèles de turbulence, à savoir la turbulence SKE et $SST k - \omega$, et la taille du pas de temps est étudié, et les résultats obtenus sont comparés avec des résultats expérimentaux.

L'objectif de cette recherche est de développer une configuration numérique pour laquelle certaines caractéristiques qualitatives et quantitatives de l'écoulement du fluide cavitant telles que la mesure du coefficient de pression (C_p), les fluctuations de pression soudaines et leurs fréquences, l'initiation de la cavitation, la croissance et son effondrement sur le profil NACA66 sont prédits. En accord avec les résultats expérimentaux et sur la base des prédictions de C_p autour du profil, la cavitation à $\sigma = 1.25$ était instable, et la cavitation associée à un écoulement cavitant avec $\sigma = 1.41, 1.34$ et 1.30 est rapportée comme stable et les longueurs de cavitation ont été prédites avec une précision décente variant de 5% à 14% d'erreur. De plus, les caractéristiques instables de la cavitation autour du profil NACA66 connue comme de la cavitation en nuages sont étudiées dans la recherche actuelle. Dans le chapitre 4, en effectuant des simulations transitoires, les prédictions de fréquences associées aux fluctuations soudaines de pression dans les écoulements cavitants à

différents nombres de cavitation, à savoir $\sigma = 1.41, 1.34$, et 1.30 , sont également démontrées en bon accord avec les résultats expérimentaux avec seulement 3,5%, 6,3 % et 10,0 % d'erreur, respectivement. Il convient de noter qu'un maillage généré à partir de près d'un million de nœuds ainsi que la taille de pas de temps la plus fine possible, soit 0,00002 (s), est utilisé pour exécuter les simulations, et le modèle de turbulence SKE et la cavitation Kubota sont utilisés pour les calculs numériques. De plus, en accord avec les résultats expérimentaux, les résultats CFD ont confirmé que la fréquence mentionnée diminue à mesure que la longueur de cavitation augmente.

ABSTRACT

Due to the complexity of cavitating flow field and sudden pressure fluctuations around the blades of hydraulic machinery, cavitation modeling is a very challenging problem, and since it has many detrimental effects on many industries such as the hydraulic industry, it has been a very interesting topic for researchers in the last decades. Since turbulence variations near hydrofoil can result in under prediction of local pressure, pressures lower than the saturation pressure might be predicted. Hence, choosing the right turbulence model which models viscosity appropriately, is of great importance.

The objective of this research is to develop a numerical setup by which some qualitative and quantitative characteristics of the cavitating fluid flow such as pressure coefficient (C_p) measurement, sudden pressure fluctuations and the frequencies associated with the highest-pressure variations, cavitation initiation, growth, and collapse over the NACA66 hydrofoil are predicted. In agreement with the experimental results and based on the C_p predictions around the hydrofoil, the cavitation at $\sigma = 1.25$ was unstable, and the cavitation associated with a cavitating flow $\sigma = 1.41, 1.34$ and 1.30 are reported to be stable and the cavitation lengths were predicted with a decent precision varying from 5% to 14% error. Furthermore, the unsteady characteristics of cavitation around the NACA66 hydrofoil experiencing cloud cavitation is studied in the current research.

The current research is performed using the ANSYS-Fluent 16.1 commercial solver and it is devoted to model cavitating flows at different Thoma numbers (σ) around a NACA66 hydrofoil. In this regard, in chapter 3, the Zwart-Gerber-Belamri cavitation model coupled with the mixed multiphase modeling approach as well as the standard $k - \varepsilon$ (SKE) turbulence model is applied, and the precision of the performed numerical predictions is investigated using the named models. Then, the effect of some other parameters on the steady-state simulation such as mesh density, convergence criteria, various turbulence models, namely SKE and *SST* $k - \omega$ turbulence, and time-step size is studied, and the obtained results are compared with the experimental results.

In chapter 4, by performing transient simulations, frequency predictions associated with the sudden pressure fluctuations in the cavitating flows at different cavitation numbers, namely $\sigma = 1.41, 1.34$, and 1.30 , are also demonstrated good agreement with the experimental results with only

3.5%, 6.3%, and 10.0% error, respectively. It should be noted that a mesh generated from near one million nodes as well as the finest feasible time-step size meaning 0.00002 (s), is used to run the simulations, and the SKE turbulence model and Kubota cavitation are employed for the numerical calculations. Furthermore, in agreement with the experimental results, the CFD results confirmed that the mentioned frequency reduces as cavitation length increases.

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LIST OF SYMBOLS AND ABBREVIATIONS

This list presents the symbols and abbreviations used in the thesis or dissertation in alphabetical order, along with their meanings.

Symbols

Y^+ Dimensionless distance from the wall

τ_w Wall shear stress

k Turbulent kinetic energy (m^2/s^2)

ω Specific turbulent dissipation rate ($1/s$)

ε Turbulent dissipation rate (m^2/s^3)

σ Thoma number

σ Surface tension (N/M)

ρ Density

ν Kinematic viscosity ($\nu = \mu/\rho, m^2/s$)

μ Mixture viscosity ($Pa \cdot s$)

μ_t Turbulent viscosity ($Pa \cdot s$)

p_{sat} Saturated vapor pressure (Pa)

p_{in} Inlet pressure (Pa)

p_{out} Outlet pressure (Pa)

α Angle of attack ($^\circ$)

α_v Vapor volume fraction (VVF)

f Frequency (Hz)

C_p Pressure coefficient

Abbreviations

AR	Aspect Ratio
BSL	Baseline
CFD	Computational Fluid Dynamics
CFL	Courant–Friedrichs–Lewy (Courant Number)
DES	Detached-Eddy Simulations
DNS	Direct Numerical Simulations
LE	Leading Edge
LES	Large Eddy Simulations
M	Mach Number
NACA	National Advisory Committee for Aeronautics
RANS	Reynolds-Averaged Navier-Stokes
Re	Reynolds Number ($Re = \frac{u_{\infty} c}{\nu}$)
RNG	Re-Normalisation Group
RSM	Reynolds Stress Model
SAS	Scale-Adaptive Simulation
SKE	Standard $k - \varepsilon$
SST	Shear Stress Transport
TE	Trailing Edge
URANS	Unsteady Reynolds-Averaged Navier-Stokes

CHAPTER 1 INTRODUCTION

1.1 Motivation

The present research is an investigation of the cavitation phenomenon using numerical method and study of frequencies associated with the sudden pressure changes in cavitating flows at various operating conditions around a two-dimensional geometry of the NACA66 hydrofoil using the commercial ANSYS FLUENT 16.1 commercial solver.

Cavitation occurs because of developing vapor or gas cavities within the operating liquid when the liquid pressure drops below the vapor pressure because of the flow acceleration without any significant change in temperature. In the water phase diagram (Figure 1.1), the transition of liquid to vapor is called boiling when it occurs by heating up the liquid and it is called cavitation whenever it is obtained by decreasing the pressure of the liquid at a constant temperature.

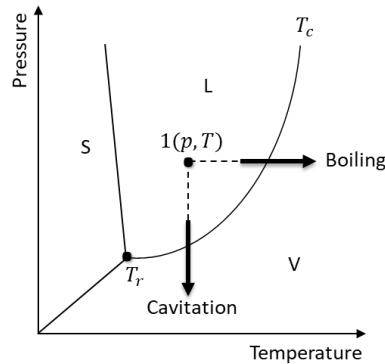


Figure 1.1 Phase diagram of water with the phase change curves revealing the boiling and cavitation phenomena

As cavitation directly impacts the hydrodynamic performance of the machine, and it may be observed in a variety of systems such as hydraulic structures, marine propellers and foils, and diesel injectors, it is crucial to predict this phenomenon.

The transition of cavitating flow from sheet cavitation to cloud cavitation is related to the turbulence in the flow. Hence, appropriately modeling turbulence, which can predict the flow details during the cavitation phenomenon, is of great importance.

The unsteadiness of the flow and the origin of cavitation shedding phenomenon is a serious challenge in simulating cavitating flows. In order to have a successful simulation to help us predict and prevent potential massive damages on the blades of hydraulic turbines, and avoid occurrence of the resonance phenomenon, it is crucial that we have a simulation in which we could appropriately model an unsteady cavitating flow.

1.2 Basic definitions and concepts

According to the flow pattern around the investigating hydraulic structure, each of the two main types of cavitation introduced in the Figure 1.2, namely the attached cavitation or convected cavitation might occur. The main forms of attached cavitation are leading edge and tip vortex cavitation. On the other hand, the bubble cavitation and convected vortex cavitation or Von-Karman street are the main types of the convected cavitation [1].

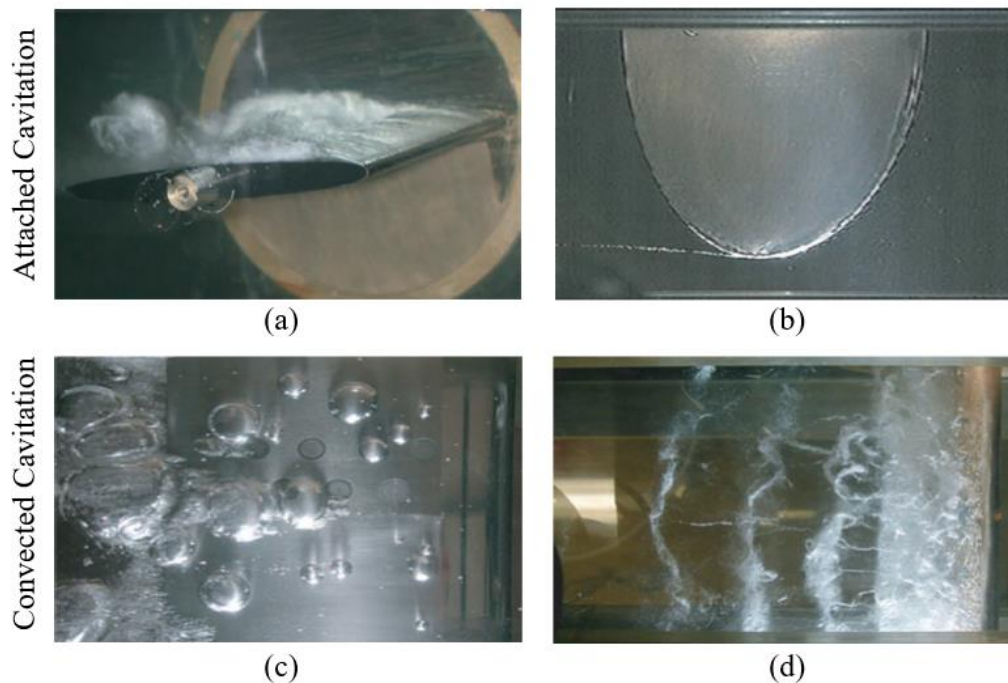


Figure 1.2 The (a) leading edge cavitation, (b) tip vortex cavitation, (c) bubble cavitation, and (d) convected vortex cavitation type in water tunnel flowing from right hand side [1].

The attached cavitation occurs when the interface of the cavity is partly attached to the body. However, the convected cavitation occurs when the whole interface is shifting with the fluid flow. The leading-edge or sheet cavitation is frequently detected in hydraulic turbines. It is characterized by a partial vapor cavity that detaches from the leading edge and it is well-known for its harsh erosion.

Convected or cloud cavitation is a type of cavitation instability which typically appears with an unsteady nature, in which large parts of the cavity are shed from a central cavity and appears like clouds in the cavity wake [2]. A growth-collapse cycle occurs in case of this cavitation type. It starts by growing a cloud of vapor bubbles in the low-pressure region, which is caused by an increasing velocity of the flow, typically near the leading edge of the hydrofoil. Subsequently, these bubbles collapse in flow regimes where the pressure recovers. The frequency of cavitation shedding depends on the cavitation number, freestream velocity, roughness of the walls, and temperature. In order to detect the transition from cloud cavitation to sheet cavitation and progress of the erosion due to cavitation, Ylonen et al [3], in an experimental study, measured the frequency of these vapor collapses using the acoustic emission (AE) sensors. They reported that with increasing cavitation number (σ), the frequency of the cavity-shedding phenomenon decreases. On the other hand, augmentation of erosion was reported as the frequency of cavitation shedding phenomenon increased. Generally, detection of periodic bubble cloud shedding in various hydraulic systems and particularly around hydrofoils is common.

The attached tip vortex cavitation typically occurs at the edges of blades, and it is represented by high shear and low-pressure fields.

The bubble cavitation occurs for hydrofoils placed in flows with low incidence angles and when single bubbles are generated and move with the fluid as they develop and collapse.

The transition of sheet cavitation to cloud cavitation is characterized by a re-entrant jet at the ending of the cavity (Figure 1.3). It is believed that the re-entrant jet is generated by development of the flow in the closure region in the wake of the cavity, moving with the wall and forming a local stagnation point. The shearing generated by the re-entrant jet is the main source of the shape variation of the bubble tail. Furthermore, the liquid jet describes as a consistent jet advancing from the closure region and it fills the cavity void (Figure 1.4).

Cavitation inception	Sheet cavitation	Pulsating cavity	Cloud cavitation
 • Apparition of vapor “fingers” and “patches” • Low dynamic behavior • Low compressibility effects • Dispersed cavities	 • Stable vapor cavity • Dynamic behavior located in the closure region • Low compressibility effects • Principally 2-D	 • Self-oscillating behavior of whole cavity • Highly dynamic • Compressibility effects in shed clouds of bubbles • Important 3-D effects	 • Chaotic behavior of cavity releases • Extremely dynamic • Compressibility through shock-waves • Important 3-D effects

Figure 1.3 Transition of cavitation type around a two-dimensional hydrofoil, revealing that reducing σ or rising AoA would increase the unsteady behaviour of the flow [4].

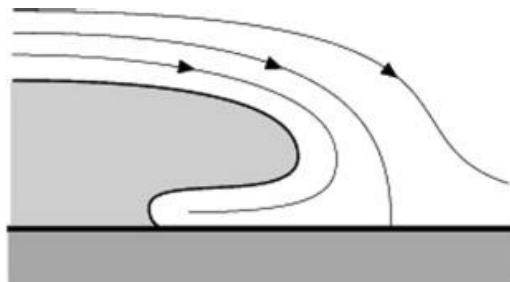


Figure 1.4 Classical model representing the re-entrant jet mechanism [5]

When the re-entrant jet reaches the stagnation point, the cavity collapses at cavity closure and its detached parts develop a vapor cloud. The named process is illustrated for a single bubble (Figure 1.5). It is remarkable that the erosion mostly occurs in the vicinity of the closure region.

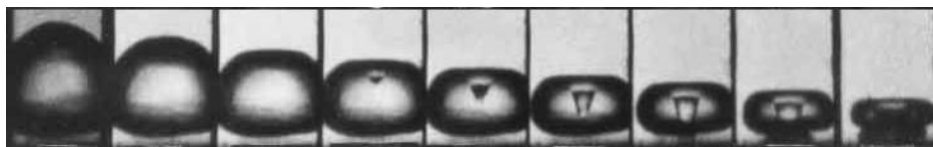


Figure 1.5 Illustration of collapse of a single bubble and microjet [6]

Arndt et al. [7] investigated the shedding behavior of the cloud cavitation at different operating conditions and concluded that the re-entrant jet appears at high values of $\sigma/2\alpha$. Whereas the bubbly flow shock wave occurs at low values of $\sigma/2\alpha$. The $(\sigma/2\alpha)$ term is a dimensionless parameter that comprises pressure and geometrical characteristics, σ is cavitation number and α is the incidence angle. The analysis demonstrated that the frequency of cavitation instabilities depends on this term. They also reported some models for different regimes of the attached cavitation. For $6 < \sigma/2\alpha < 8.5$, small vapor, called patches appeared around the hydrofoil. The unstable cavitation which has extremely dynamic nature, caused by re-entrant jet was reported for $4 < \sigma/2\alpha < 6$. Ultimately, at the range of $1 < \sigma/2\alpha < 4$, transition of cavitation type to the cloud cavitation led to occurrence of shockwaves in the flow-field.

From experiments it was observed that the cavity length decreased with increasing cavitation number. Furthermore, it is observed that the shedding often occurred when the cavity was separated from the leading edge. When a big bubble is shed, the pressure fluctuation due to its subsequent collapse suppress the expansion of the recently developing partial cavities. This phenomenon causes cavity shedding in different steps which is destructive in the hydraulic machines and leads to efficiency loss.

1.3 Present work

In the last decades several numerical simulations were performed to investigate the different characteristics of cavitation phenomenon. Several numerical predictions were accomplished using various solvers such as ANSYS Fluent, ANSYS CFX [8], and OpenFOAM open source solver. The present study aims to numerically predict frequency of cavity-shedding around the NACA66 hydrofoil with two-dimensional geometry, using the ANSYS Fluent solver. The results of this study will be helpful analysis guidelines for hydraulic engineers, looking for an efficient and robust approach to predict and prevent erosion and resonance phenomenon in hydraulic machines.

The main goal of this research is to find adequate simulation parameters by which the results are in a good agreement with the experimental results presented by Leroux et al [9]. Then, a turbulence model should be selected in a way that allows capturing most of the complexities of the flow for varieties of cavitating condition.

Furthermore, the following specific objectives are going to be the main concentration of the current research:

- Develop a steady-state method to perform numerical simulation for the fluid flow under non-cavitating and cavitating condition using the best tested numerical parameters.
- Developing an unsteady method to study characteristics of the cavitating flow qualitatively and quantitatively and investigate cavitation behaviour for the cavitating flows at various cavitation numbers.

Based on the results of the simulation of cavitating flow over a three-dimensional rectangular hydrofoil reported by Wang et al [10], since the flow is uniform in almost 90% of the span surface of the hydrofoil, it is decided to perform the simulations on a two-dimensional hydrofoil to enhance the efficiency of computations.

1.4 Thesis outline

The introduction and literature review sections present the research objectives and the latest advances in the field. This thesis consists of three more chapters.

Chapter 3 presents the Methodology of the current research by which the steady-state simulations on the named case studies are intended to be performed. Furthermore, the effect of several numerical parameters on the precision of our predictions such as generating denser meshes and using different turbulence models, and impact of convergence criteria on the solution are investigated and the predictions are compared with the experimental results. Then, the cavitation length in the flows at various cavitation numbers are calculated and compared with the experimentally measured values and the error associated with the employed method is calculated in this chapter.

chapter 4 is devoted to investigating the unsteady behaviour of the cavitating flow, qualitatively and quantitatively. Then, the frequencies associated with the pressure fluctuation intensity at the cavity closure in cavitating flows at different cavitation numbers are calculated. In this regard, three time-step sizes were selected, and the impact of the selected time-step sizes were investigated on the precision of the predicted frequencies. Then, some conclusions are drawn based on the validation of the numerical results with the experimental results.

Finally, the key accomplishments and the main contribution of the current work as well as its limitations and some recommendations for the future studies to answer more questions in the field are stated in chapter 5.

CHAPTER 2 LITERATURE REVIEW

To select the parameters based on which the numerical simulations will be calculated, the latest advances in the field are studied. The present review is written to determine a sensible approach for performing the simulations as well as defining certain terms. We start this chapter by presenting the necessary constitutive equations, boundary conditions and the main categories of the computational grids and their characteristics. Then, we continue by introducing other important parameters that should be considered for steady and unsteady simulations of the cavitation phenomenon such as homogeneous and inhomogeneous models, discretization in time and space, turbulence models, eddy viscosity modification and cavitation models.

2.1 Constitutive equations

This chapter starts by defining the usefull equations in the current research. It is important to mention that for simplification of the simulations, we have considered the fluid to be incompressible, where the density of the operating fluid to be constant. It is noteable that since the cavitation is considered as an isothermal phenomenon, the conservation of energy equation is not solved in our simulations and the governing equations are the Navier-Stokes equations with some additional equations related to the turbulence closure from the selected turbulence model, and a transport equation to solve the vapor mass fraction, and momentum and continuity equations. The continuity and momentum equations for the unsteady flow are stated in the following:

$$\frac{\partial}{\partial t}(\rho_m) + \frac{\partial}{\partial x_j}(\rho_m u_j) = 0 \quad (1)$$

$$\begin{cases} \frac{\partial}{\partial t}(\rho_m u_i) + \frac{\partial}{\partial x_j}(\rho_m u_i u_j) = -\frac{dP}{dx_i} + \frac{d\tau_{ij}}{dx_j} \\ \tau_{ij} = (\mu + \mu_t) \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \delta_{ij} \end{cases} \quad (2)$$

Where, u is the velocity vector, P is pressure, μ and μ_t are viscosity and turbulent viscosity, δ_{ij} is the Kronecker delta, ρ_m is the density of mixture and it will be calculated using the following equation:

$$\rho_m = \alpha_v \rho_v + (1 - \alpha_v) \rho_l$$

Where, the subscripts l and v represent the liquid, and vapor phase.

These equations will be discretized using the schemes available in the solver which are mentioned later in this chapter.

In the present work, the velocity and pressure are calculated using the ANSYS-Fluent 16.1 commercial software. All calculations are performed based on the steady and unsteady Reynolds Averaged Navier-Stokes, respectively RANS and URANS. The RANS equations are mentioned as follows:

$$\rho \frac{\partial U_i}{\partial t} + \rho \frac{\partial (U_i U_j)}{\partial x_j} = - \frac{\partial P}{\partial x_i} + f + \mu \left(\frac{\partial^2 U_i}{\partial x_j^2} \right) + \rho \frac{\partial}{\partial x_j} \overline{u'_i u'_j} \quad (3)$$

Where, U is the velocity time-average, P is the average pressure, f is a body force per unit mass of fluid, μ is the dynamic viscosity, and $\rho \overline{u'_i u'_j}$ is the Reynolds shear stress. Based on the Boussinesq hypothesis[11], the Reynolds shear stress may be written as:

$$\rho \overline{u'_i u'_j} = -\mu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \frac{2}{3} \rho \delta_{ij} k \quad (4)$$

Where, μ_t is turbulent viscosity, δ_{ij} is the Kronecker delta, and $k = \frac{1}{2} \sqrt{\overline{u'_i u'_j}}$. The various parameters that arise from the analysis are discussed in the following section.

2.1.1 Reynolds number

The Reynolds number (Re) in fluid mechanics is introduced in a dimensionalization of the equations and represents the relative importance of the viscous forces over the inertial forces. It is therefore an important dimensionless parameter which characterizes the flow condition. It determines whether the fluid flow is laminar or turbulent. The transition between laminar and turbulent flow occurs in a range starting between 1000 and 2000 and extending to between 3000 and 5000. Generally, when the value of the Reynolds number is less than about 2000, fluid flow is mostly laminar.

The Reynolds number in the vicinity of a hydrofoil should be calculated using the following equation:

$$Re = \frac{\rho u_{\infty} c}{\mu} = \frac{u_{\infty} c}{\nu} \quad (5)$$

Where, ρ is the fluid density, μ is the molecular viscosity, ν is the kinematic viscosity of the fluid ($\nu = \mu/\rho$), u_{∞} is the free-stream velocity, and c is the chord length of the hydrofoil.

In hydraulic turbine applications, the order of magnitude of the Reynolds number varies from 10^5 to 10^8 , based on the complexity of the geometry [12], and the flow is therefore always considered turbulent. Hence, implementation of a suitable turbulence model to study the flow is necessary.

2.1.2 Thoma number

Cavitation is described by a non-dimensional number, named the Thoma number or cavitation parameter (σ).

$$\sigma = \frac{p_{ref} - p_v}{\frac{1}{2} \rho u_{\infty}^2} \quad (6)$$

Where, p_v and p_{ref} are the vapor pressure and outlet pressure, respectively, and ρ and u_{∞} are the density and free-stream velocity of the fluid. Based on the equation (6), the lower the Thoma

number, the closer p_v and p_{ref} are going to be. Consequently, the lower the Thoma number is, the more probable bubble formation and cavitation are to occur.

Delgosha et al [13], in a two-dimensional CFD study on the NACA0015 foil with an angle of attack of 7° , concluded that if the cavitation number is greater than 3.5, non-cavitating condition occurs. Then, they reported that for a cavitation number in range of $2 < \sigma < 3.5$, almost steady sheet cavitation occurs. Furthermore, unsteady cavitation with periodic cloud shedding behavior, and severe cavitating condition close to super cavitation were seen for cavitation numbers in the range of $0.8 < \sigma < 1.7$ and $\sigma = 0.5$, respectively. It is noteworthy that the super cavitation is the use of a cavitation bubble to decrease skin friction drag on a flooded object and eventually it allows high speeds. It might occur in a variety of applications such as torpedoes and propellers, and theoretically a whole submerged vessel.

2.2 Discretisation schemes

Different codes offer various discretization schemes to the users to implement the simulation. Most CFD codes, including the ANSYS FLUENT commercial solver, calculate the solution and store the information of the flow (ϕ_p) at the centroid of each cell in the computational grid. Though, these values (ϕ_f) are needed to be known on every cell faces. Each code calculates the values on the cell faces using various interpolation schemes.

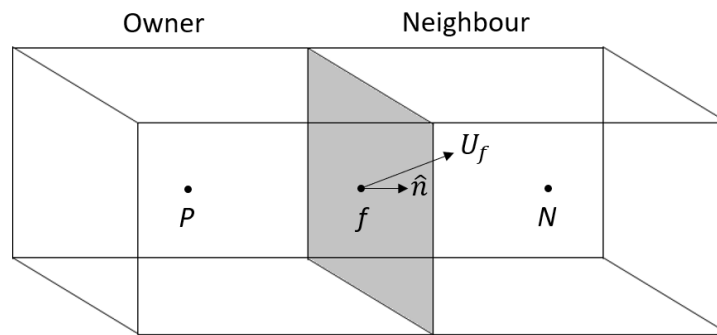


Figure 2.1 The control volumes in a hexahedral computational grid, revealing the cell centers and the selected internal face-cell

One of the most popular form of schemes is based on upwind differencing that is used for the convection term. This type of differencing depends on the direction of the mass flow flux, streaming through the cell [14].

$$F_f = \rho_f A_f (U_f \cdot \hat{n}) \quad (7)$$

Where, ρ_f is the density, A_f is the area, and the U_f is the velocity of the fluid on the face-cell at which the value should be calculated.

If the value of the mass flux (F_f) is positive, it indicates that the flow is streaming from the owner cell into the neighbor cell, and the negative mass flux means that the flow is streaming into the owner cell. Then, the value of ϕ_f is provided by:

$$\phi_f = \begin{cases} \phi_P & F_f > 0 \\ \phi_N & F_f < 0 \end{cases} \quad (8)$$

It is notable that as the value of ϕ in the upwind differencing method does not change linearly through the cell-face and its value is constant between the cell centroid and cell-face, this method is first-order accurate and thus not very precise. However, since it is the most stable scheme for convection dominated flows, most CFD codes include some term of upwind differencing.

The existing discretization schemes for the convection term to interpolate the values of the flow to the cell faces in the FLUENT solver are First-Order Upwind, Power Law, Second-Order Upwind, Monotone Upstream-Centered Schemes for Convection Laws (MUSCL), and Quadratic Upwind Interpolation (QUICK) schemes [15]. The Upwind schemes are known as the easiest schemes to converge. Though, the Second-Order Upwind scheme uses larger stencils to obtain the second order of accuracy and the convergence would be a little slower than the First-Order Upwind scheme. For the flows with low Reynolds number ($Re_{cell} < 5$), the results obtained with the Power Law scheme would be more precise in comparison with the ones acquired with the First-Order scheme. The MUSCL scheme is a 3rd order discretization scheme for unstructured meshes and it predicts the forces, vortices, and the secondary streams well. The QUICK scheme has 3rd order of

accuracy on uniform meshes, and it operates with the quadrilateral, hexahedron, and hybrid computational grids.

Furthermore, the Standard, PRESTO, Linear, Second Order, and Body Force Weighted schemes are available in the pressure-based solver in ANSYS Fluent to calculate pressure values at the cell-faces [15]. The Standard scheme is the default scheme in the solver. This scheme provides reduced accuracy for the flows exposing significant pressure gradients near the boundaries. Hence, it should not be applicable since we expect to have steep pressure gradients. A decent alternative for modeling flows with high pressure gradient such as highly swirling streams is the PRESTO. scheme. Then, if the simulation does not converge well, the Linear scheme is a good option. On the other hand, the Second-Order scheme is habitually proposed for modeling compressible flow, and it is not recommended for simulations with porous media or formulations involving VOF or mixture multiphase models. Lastly, the Body Force Weighted scheme is a good scheme for problems in which large body forces or highly swirling flows are expected.

To model the phenomena in which a strong inter-dependence exists among density, energy, momentum, and species, coupled solvers are recommended. Li et al [16] selected the coupled implicit solver to resolve the momentum and pressure equations to have a fast convergence and perform a robust computation. They applied the QUICK discretization scheme in space, PRESTO scheme to compute pressure on the cell-faces, and First-Order implicit in time, to investigate the reliability of the prediction of cavitation erosion using the pressure-based ANSYS FLUENT commercial software.

2.3 Multiphase modeling approaches

The existing approaches for numerical modeling of multiphase flows include the Eulerian-Lagrangian (EL) and the Eulerian-Eulerian (EE) approaches. There are also two two-phase interaction models, namely dispersed-continuous and continuous-continuous. Dispersed phase is usually characterized by small pocket diameters from μm to mm , which are dissolved in the continuous phase. These pockets could be particles, droplets, or bubbles. The dispersed-continuous phase interaction models describe the interaction between the dissolved phase and the continuous

phase. Sprays, air conditioning, and sediment transport applications are some typical applications of the dispersed-continuous phase interaction models. It is noteworthy that the dispersed-continuous phase interaction occurs in the cavitation phenomenon and hence it could potentially be an appropriate model for cavitation modeling.

On the other hand, the continuous-continuous phase interaction occurs when two phases form a discrete interface between them. At these interfaces, the phases are usually immiscible with each other. For instance, the phase interaction in surface wave modeling applications and all other applications in which we care about tracking the interface or free surface are best described using continuous-continuous models.

The Eulerian multi-phase models may resolve dispersed-continuous and continuous-continuous phase interactions. In the Eulerian-Lagrangian approach, the 2D motion of every element is computed from the existing forces, accounting for the interaction among the fluid phases. Furthermore, the volume fraction of the dispersed phase must be neglected in this approach. Since volume fraction of the water vapor is important to model cavitation, the EL approach is not the best approach in the current research. Various, the Eulerian-Eulerian approach treats different phases as continuous compressible flow, and the volume fraction is defined for each phase. Hence, the EE approach is the desirable approach to model the cavitation phenomenon.

The mixture and volume of fluid (VOF) models are simplified versions of the full Eulerian model for the dispersed-continuous and continuous phase interactions, respectively.

A Eulerian model solves the continuity or mass-conservation equation for each phase in the system. However, in the mixture and VOF models, one single continuity equation needs to be solved.

The modified continuity equation for the phase q is stated in the equation (9), where r_q is the volume fraction of the phase q , ρ is density of the fluid, U is the free-stream velocity, and $\dot{m}_{pq}$ and $\dot{m}_{qp}$ are the mass transfer between phases of p and q . Moreover, the mass transfer rates describe the phase change processes.

$$\frac{\partial(r_q \rho_q)}{\partial t} + \nabla \cdot (r_q \rho_q U_q) = \sum_{p=1}^N (\dot{m}_{pq} - \dot{m}_{qp}) \quad (9)$$

If we have dispersed-continuous phase interaction in our simulation using ANSYS Fluent solver, the volume fraction should be able to take any value between 0 and 1, meaning that for the cells in which the value of volume fraction is zero, there would be no dispersed phase. Then, when the value of volume fraction increases in certain cells, it reveals that the concentration of the dispersed phase in those cells is increasing. On the other hand, in the continuous-continuous phase interaction, on one side the value of the volume fraction is expected to have a value of zero and on the other side of the interface, normally in the interface region where the dispersed phase exists, the value of volume fraction should be one.

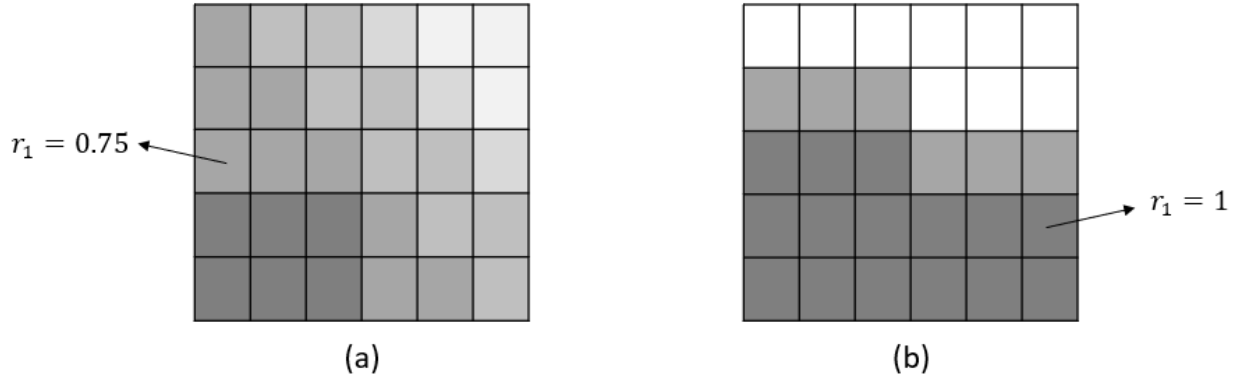


Figure 2.2 The approach of CFD solver to distinguish between the (a) dispersed-continuous and (b) continuous-continuous phase interaction

The CFD code solves the volume fraction equation, which is stated in the following:

$$\frac{\partial r_q}{\partial t} + \nabla \cdot (r_q U_q) = 0 \quad (10)$$

Since the sum of volume fractions must be equal to one, the transport equation should be resolved $N - 1$ times. Where, N is the number of phases in our fluid flow. This approach is identical to Eulerian, mixture and VOF models.

The CFD code distinguishes between the two types of phase interaction using a special discretization scheme such as Compressive Interface Capturing (CICSAM) and Piecewise-Linear

Interface Construction (PLIC) instead of the upwind differencing, on the convection term of $\nabla \cdot (r_q U_q)$ in the equation above.

Indeed, numerically solver enforces a sharp interface between the two phases using the above-named schemes. This approach is named the interface capturing approach and it is applied in the full Eulerian and VOF models. Although, since we have dispersed-continuous phase interaction in the mixture model, this approach is not used in the mixture model.

Three multiphase models, namely VOF, mixture, and Eulerian models are available and could be enabled in the ANSYS Fluent 16.1 solver. It is noteworthy that if the VOF and Eulerian models are selected, the volume fraction scheme must also be specified along with the number of phases. Then, the phase interactions should be defined; the surface tension phase interaction for the VOF model, slip velocity functions in case of mixture model selection, and drag functions phase interaction should be selected if the Eulerian model is employed in the simulation.

Based on characteristics mentioned for the available models, the mixture model is the most appropriate approach by which the cavitation could be modeled precisely and efficiently in the current case studies, and it is chosen to perform the simulations.

2.4 Turbulence model

Turbulence is a complicated process that exhibits fluctuations in time and space. The reason of this complexity is related to the fact that it involves several scales and many fluid characteristics, while being also three dimensional and unsteady. To introduce the notion of turbulence flow modeling we only concentrate on incompressible single-phase flow with constant laminar viscosity. Habitually, it is described by eddy viscosity as a local property of the fluid. Depending on the fluid velocity, the Reynolds number in the problems in which an item is immersed in a fluid is high, resulting in a medium to highly turbulent flow, like the current research. Hence, to model the fluid flow around a hydrofoil, it is necessary to choose a proper turbulence model.

Currently, there exist three main approaches to simulate turbulence and each one represents a certain degree of resolution. These approaches could be classified as Reynolds Averaged Navier-Stokes (RANS), Large Eddy Simulation (LES), and Direct Numerical Simulations (DNS). These

approaches differ mainly in the extent of schemes being resolved while solving the Navier-Stokes equations.

2.4.1 Reynolds Averaged Navier Stokes (RANS) model

The traditional linear eddy viscosity RANS models could be divided into the following groups [17]: algebraic (zero equation) models, half equation models, one equation models, and two equation models. The algebraic, one and two equation models are available in the commercial ANSYS Fluent solver, and they will be discussed in a subsequent section, after presenting the following two simulation models.

The RANS simulation approach applies an averaging procedure to the Navier-Stokes equations. The URANS model, uses the same approach to model flow for unsteady problems.

The Reynolds-Averaged Navier-Stokes (RANS) and unsteady Reynolds-Averaged Navier-Stokes (URANS) equations are applied to model the stationary attached pocket cavitation, and unsteady sheet and cloud cavitation, respectively. The conservative form of the URANS equations for a Newtonian fluid is introduced in the following.

The mass conservation equation is:

$$\frac{\partial \rho_m}{\partial t} + \frac{\partial (\rho_m u_j)}{\partial x_j} = 0 \quad (11)$$

Conservation of momentum equation:

$$\frac{\partial (\rho_m u_i)}{\partial t} + \frac{\partial (\rho_m u_i u_j)}{\partial x_j} = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} [(\mu_m + \mu_T) \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right)] \quad (12)$$

The mass transfer equations between the phases:

$$\frac{\partial \rho_l \alpha_l}{\partial t} + \frac{\partial (\rho_l \alpha_l u_j)}{\partial x_j} = \dot{m}^+ + \dot{m}^- \quad (13)$$

The equation for the density of the mixture:

$$\rho_m = \rho_l \alpha_l + \rho_v \alpha_v \quad (14)$$

The mixture viscosity equation:

$$\mu_m = \mu_l \alpha_l + \mu_v \alpha_v \quad (15)$$

Where, u and p are the velocity and pressure of the fluid, ρ_m , ρ_l and ρ_v are the density of mixture, liquid, and vapor, respectively, μ_T is the turbulent viscosity, α_v and α_l are the volume fraction of vapor and liquid, respectively, μ_m , μ_l and μ_v are the dynamic viscosity of mixture, liquid and vapor. The source terms of $\dot{m}^+$ and $\dot{m}^-$ depend on the chosen cavitation model and they represent the mass transfer rate between phases per unit volume for the condensation and evaporation phenomenon, respectively.

The turbulence models available in the commercial ANSYS Fluent software are listed in the following table.

Table 2.1 The turbulence models available on the ANSYS Fluent commercial solver

Model	Number of eqn.	Simulation Approach	Details
Spalart-Allmaras (SA)	1	RANS	Intended mostly for aerospace applications. It is effective for low-Reynolds number model. It provides decent results for boundary layers subjected to adverse pressure gradients [18].
Standard $k - \varepsilon$ (SKE)	2	RANS	It uses wall function ($30 < Y^+ < 100$), C_μ is constant, and it is typically more stable than the RNG and Realizable models [19].
$k - \varepsilon$ RNG	2	RANS	Re-Normalization Group (RNG). It introduces a modified format for the ε equation. The Navier-Stokes equations were renormalized to account for the effect of smaller scales of motion. It uses wall function ($30 < Y^+ < 100$) [20].
$k - \varepsilon$ Realizable	2	RANS	It uses wall function ($30 < Y^+ < 100$), it has a new expression for the turbulence viscosity, C_μ is a variable, new transport equation for the turbulence dissipation rate (ε), it provides enhanced estimates for the spread rate of jets.

Model	Number of eqn.	Simulation Approach	Details
Standard $k - \omega$	2	RANS	Two-equation model (turbulent kinetic energy and specific dissipation rate), k determines the energy of turbulence and ω determines the scale of turbulence. It is very sensitive to the inlet free-stream turbulence properties [21].
BSL $k - \omega$	2	RANS	New baseline (BSL) model works like the Wilcox's standard $k - \omega$ model with less dependency on the freestream values.
SST $k - \omega$	2	RANS	It was proposed by Menter in 1994 [22], it can be used in flows with low-Reynolds, SST formulation switches to a $k - \varepsilon$ behavior in the free-stream (it avoids common problems with the $k - \omega$ model).
Transition $k - k_l - \omega$	3	RANS	It is also known as eddy viscosity model, it was proposed by Walters and Cokljat (2008), for adverse pressure gradient flows, it solves equations for laminar/kinetic energy (k and k_l), and time scale $\omega = \varepsilon/k$.
Transition SST	4	RANS	It is based on the coupling of the SST $k - \omega$ transport equations with one more equation for intermittency and the other one for the transition inception criteria in terms of momentum-thickness Reynolds number. It is developed to cover standard transition as well as the flows in turbulent conditions [23].
Reynolds Stress (RSM)	5	RANS	It originates from the work of Chou [24] and Rotta [25]. It is the most comprehensive classical turbulence model. The second order closure method is used. On this model, the eddy viscosity approach is avoided, and particular elements of the Reynolds stress tensor are directly computed [26].
Scale-Adaptive Simulation (SAS)	2	Hybrid (LES and RANS)	It was proposed by Menter and Egorov, it is recognized as a powerful model for unsteady simulations. It expands URANS to several flows.
Detached Eddy Simulation (DES)	2	Hybrid (LES and RANS)	It was proposed first by Spalart et al in 1997, it was designed for simulations with high Reynolds number, coarser mesh than LES is needed, more complicated mesh generation than RANS and LES [27].

It is noteworthy that by increasing the number of involved equations going down the list in Table 2.1, due to an increasing complexity of the turbulence models, the calculation cost increases gradually.

According to the number of equations in the mentioned turbulence models, one equation model is categorized by formulating one supplementary transport equation to calculate turbulence quantity and that is normally the turbulence kinetic energy (k). Since for all models we need to prescribe a length-scale distribution based on the experimental results, and that for elliptic flows such as separated and recirculating flows, the experimental results are not available, it is challenging to prescribe length scale and that is why most researchers prefer to adopt models with two or more equations [28]. These models are often suitable for simulating flow on the industries with aeronautics and nuclear applications.

The two-equation models, use two transport equations besides the mean-flow Navier-Stokes equations; the first one is typically for the turbulence kinetic energy (k) and the other one is from a variety of variables that includes: the dissipation rate of turbulence kinetic energy (ε), specific dissipation rate (ω), length scale (l), time scale (τ), and the product of $k \sim l$ of k and ε [28].

The turbulent motion as a random variation around a mean value is acquired by Reynolds decomposition, defined in the reference [29] as equation (16), where ϕ is the instant scalar quantity, $\bar{\phi}$ is the time-average value, and ϕ' is the fluctuating part.

$$\phi = \bar{\phi} + \phi' \quad (16)$$

The time-average of the variable part is zero ($\bar{\phi'} = 0$) and one obtains equation (17), where t_1 is the time scale of the quick variations and t_2 is the time step of the gradual change.

$$\overline{\phi(x)} = \lim_{t \rightarrow \infty} \frac{1}{\Delta t} \int_{t_1}^{t_1 + \Delta t} \phi(x, t) dt \quad t_1 \ll \Delta t \ll t_2 \quad (17)$$

With single flow and incompressible consideration of the flow and getting the time average of these resulting equations are obtained:

$$\frac{\partial \bar{u}_i}{\partial x} = 0 \quad (18)$$

$$\frac{\partial \bar{u}_i}{\partial x} + \frac{\partial}{\partial x_j} (\bar{u}_i \bar{u}_j) = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j} (\overline{u'_i u'_j}) \quad (19)$$

where, $\bar{u}_i$ is the average of velocity, u'_i is the variable velocity, ρ is the fluid density and ν is the kinematic viscosity. The equation (19) is known as the RANS equation and the $\overline{u'_i u'_j}$ term is the Reynolds-stress tensor (τ_{ij}). Since the number of unknown quantities are larger than the number of offered equations, to solve this closure problem, the Boussinesq approximation [30] is used. Here, the turbulence eddies are considered as parts of the fluid, and they obey the kinetic theory of gasses. Hence, the turbulence stresses are developed in the equations (20, 21).

$$\tau_{ij} = \overline{u'_i u'_j} = \frac{2}{3} \kappa \delta_{ij} - \nu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \quad (20)$$

$$\kappa = \frac{1}{2} \overline{u'_i u'_i} = \frac{1}{2} (\overline{u_1'^2} + \overline{u_2'^2} + \overline{u_3'^2}) \quad (21)$$

Where, κ is turbulence kinetic energy and ν_t is eddy viscosity, which is variable and δ_{ij} is the Kronecker delta. Then, by substituting the equation (20) into the equation (19) we have equation (22).

$$\frac{\partial \bar{u}_i}{\partial t} + \frac{\partial}{\partial x_j} (\bar{u}_i \bar{u}_j) = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} [(\nu + \nu_t) \frac{\partial \bar{u}_i}{\partial x_j}] \quad (22)$$

Analysis stated that the unknown ν_t should be proportional to the product of specific velocity V_t and length scale L_t . Then, each simulation approach uses the method stated in the following to compute the formulation. The zero-equation models consider both V_t and L_t as algebraic terms. The one-equation models deem V_t as the square root of the turbulence kinetic energy ($\sqrt{\kappa}$) and L_t as the length scale.

$$\nu_t = C_v \sqrt{\kappa} L \quad (23)$$

Although, the two-equation models such as $k - \varepsilon$ and $k - \omega$ use differential equations to compute V_t and L_t [29].

$$V_t = \begin{cases} C_\mu f_\mu \frac{\kappa^2}{\varepsilon} & (k - \varepsilon \text{ model}) \\ V_t = \alpha \frac{\kappa}{\omega} & (k - \omega \text{ model}) \end{cases} \quad (24)$$

Where, f_μ is a damping function, and C_μ and α are constants, ε is turbulence energy dissipation rate and ω is dissipation per unit turbulence kinetic energy.

Among the two-equation models presented in the table above, the standard $k - \varepsilon$ and *SST* $k - \omega$ turbulence models are employed to perform the steady-state simulations. On the other hand, however several attempts were made to use *SST* $k - \omega$, Scale-Adaptive Simulation *SST* (SAS-*SST*) turbulence models to simulate cavitating flow with more precision, the transient simulations only converged when the standard $k - \varepsilon$ turbulence model was employed.

2.4.1.1 Standard $k - \varepsilon$ turbulence model

The first attempts for developing the $k - \varepsilon$ turbulence model were made by Chou in 1945[31], but this turbulence model is usually considered to have been proposed by Launder et al. in 1973 [32]. This model is often mentioned as the standard $k - \varepsilon$ model. This model is a robust and precise model for numerical simulation of engineering flows specially for the free-shear layers as well as the wake region and it is the most employed two-equation model.

The transport equations of the standard $k - \varepsilon$ model (SKE) is recapitulated in the following [30].

The transport equation of the turbulence kinetic energy (k) is:

$$\rho \frac{\partial k}{\partial t} + \rho U_j \frac{\partial k}{\partial x_j} = \tau_{ij} \frac{\partial U_i}{\partial x_j} - \rho \varepsilon + \frac{\partial}{\partial x_j} [(\mu + \mu_t / \sigma_k) \frac{\partial k}{\partial x_j}] \quad (25)$$

The transport equation of the turbulence dissipation rate (ε) is:

$$\rho \frac{\partial \varepsilon}{\partial t} + \rho U_j \frac{\partial \varepsilon}{\partial x_j} = C_{\varepsilon 1} \frac{\varepsilon}{k} \tau_{ij} \frac{\partial U_i}{\partial x_j} - C_{\varepsilon 2} \rho \frac{\varepsilon^2}{k} + \frac{\partial}{\partial x_j} [(\mu + \mu_t / \sigma_\varepsilon) \frac{\partial \varepsilon}{\partial x_j}] \quad (26)$$

And the equation of the eddy viscosity (μ_t) is:

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \quad (27)$$

Furthermore, we have:

$$\omega = \frac{\varepsilon}{C_\mu k} \quad (28)$$

$$l = \frac{C_\mu k^{3/2}}{\varepsilon} \quad (29)$$

Where, k is the turbulence kinetic energy, ε is the turbulence dissipation rate, and for the standard $k - \varepsilon$ model the defined constants are:

$$C_{\varepsilon 1} = 1.44, C_{\varepsilon 2} = 1.92, C_\mu = 0.09, \sigma_k = 1.0, \sigma_\varepsilon = 1.3$$

2.4.1.2 $k - \omega$ turbulence model

The first two-equation turbulence model proposed by Kolmogorov in 1942 [33]. He chose turbulence kinetic energy as one of his turbulence parameters and the other parameter was dissipation per unit turbulence kinetic energy (ω). The key characteristic of the $k - \omega$ model is that it provides a more accurate solutions in the near wall boundary regions.

With combination of a physical logic and dimensional study, Kolmogorov proposed the following equation for ω :

$$\rho \frac{\partial \omega}{\partial t} + \rho U_j \frac{\partial \omega}{\partial x_j} = -\beta \rho \omega^2 + \frac{\partial}{\partial x_j} \left[\sigma \mu_t \frac{\partial \omega}{\partial x_j} \right] \quad (30)$$

Where, β and σ are two new closure coefficients.

The appearance of the above equation was altered as the $k - \omega$ model developed over the past decades. Similar to Kolmogorov, Wilcox [30] wrote the equation of ω in terms of ω . However, most of the known $k - \omega$ models use an equation for ω^2 , instead.

The equation of the eddy viscosity is:

$$\mu_t = \rho \frac{k}{\omega} \quad (31)$$

The turbulence kinetic energy (k) is calculated by:

$$\rho \frac{\partial k}{\partial t} + \rho U_j \frac{\partial k}{\partial x_j} = \alpha \frac{\omega}{k} \tau_{ij} \frac{\partial U_i}{\partial x_j} - \beta \rho \omega^2 + \frac{\partial}{\partial x_j} [(\mu + \sigma^* \mu_t) \frac{\partial k}{\partial x_j}] \quad (32)$$

And the equation of the specific dissipation rate (ω) is:

$$\rho \frac{\partial \omega}{\partial t} + \rho U_j \frac{\partial \omega}{\partial x_j} = \alpha \frac{\omega}{k} \tau_{ij} \frac{\partial U_i}{\partial x_j} - \beta \rho \omega^2 + \frac{\partial}{\partial x_j} [(\mu + \sigma \mu_t) \frac{\partial \omega}{\partial x_j}] \quad (33)$$

Where:

$$\epsilon = \beta^* \omega k \quad \text{and} \quad l = \sqrt{k}/\omega$$

Where, the closure coefficients are:

$$\alpha = 5/9, \beta = 3/40, \beta^* = 0.09, \sigma = 1/2, \text{ and } \sigma^* = 1/2$$

2.4.1.3 Scale-Adaptive Simulation (SAS) turbulence model

The Scale-Adaptive Simulation (SAS-SST) turbulence model is similar to the DES model. While simpler turbulence models provide a length scale proportional to the shear layer thickness [34], these SAS-SST models adjust dynamically to the length scale of the resolved structure, and it is widely employed to model unsteady cloud cavitation. The transport equation of the turbulence dissipation is stated in the following:

$$\frac{\partial(\rho\omega)}{\partial t} + \frac{\partial(\rho\bar{U}_j\omega)}{\partial x_j} = \tilde{\alpha}\rho S^2 - \tilde{\beta}\rho\omega^2 + \frac{\partial}{\partial x_j} \left[\frac{\mu_t}{\tilde{\sigma}_\omega} \frac{\partial\omega}{\partial x_j} \right] + \frac{2\rho}{\sigma_\phi} \frac{1}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial\omega}{\partial x_j} + F_{SAS} \quad (34)$$

$$F_{SAS} = \max \left[\bar{\rho}\zeta_2 S^2 \left(\frac{l}{L_{vk}} \right)^2 - c_{SAS} \frac{2\bar{\rho}k}{\sigma_\phi} \cdot \max \left(\frac{1}{k^2} \frac{\partial k}{\partial x_1}, \frac{1}{\omega^2} \frac{\partial\omega}{\partial x_1} \frac{\partial\omega}{\partial x_1} \right); 0 \right] \quad (35)$$

Where, we have $c_{SAS} = 2$.

Although URANS method could be another option besides the SAS-SST to simulate unsteadiness of the flow, since SAS-SST model extends URANS to several technical flows and it offers more precise details in the unsteady simulations, the results calculated using SAS-SST method are much more precise in comparison with simpler models.

2.4.2 Large Eddy Simulation (LES) model

The LES approach relies on the definition of large and small scales. In this method, the small scales of the Navier-Stokes equations are filtered, and only the large scales of the Navier-Stokes are resolved [35]. Compared to the DNS method, it significantly reduces the calculation time. But since it requires very fine meshes, it is much more expensive than RANS approach and it only begins to be used for some industrial applications with simple geometries.

2.4.3 Direct Numerical Simulations (DNS) model

Between the available approaches, the DNS had a significant influence in turbulence research in the past decades [36]. The primary efforts to study homogeneous turbulence using DNS approach started at National Center for Atmospheric Research (NCAR) by Lilly [37], and Orszag and Patterson [38] in the 1970s aimed at two-dimensional and three-dimensional simulations, respectively. The DNS approach is the most precise method with high computing resources that resolves the governing equations even for the smallest elements. Indeed, the DNS method is helpful for studying of turbulent mechanisms, developments of turbulence models, and to evaluate two-

point closure systems. However, performing numerical simulations using this approach has limitations for calculating the fluid flow in the complex geometries.

2.5 Cavitation modeling

There are several cavitation models available to be used to perform cavitation modeling using CFD codes. The main difference between these models is related to the way they consider condensation and evaporation equations. There are three different cavitation models available in the FLUENT solver, namely, the Schnerr-Sauer (2001), Singhal (2002), and Zwart-Gerber-Belamri (2004) cavitation models. These models are based on the Rayleigh-Plesset equation which describes the development of a single bubble in a liquid and oversees the mass transfer from liquid phase to vapor phase. Both evaporation and condensation equations are considered in every available cavitation models. The physical basics and mathematical interpretations of the source terms of some homogeneous cavitation models are presented in the following sections.

2.5.1 Homogeneous cavitation models

The numerical methods for simulating the cavitation phenomenon are divided in two major categories. The first group is based on the interphase dynamics and the other group considers the fluid as a multiphase mixture which has an average density. The simplifications, considered for the named cavitation models are (1) using a formula which describes the bubble radius change based on the Plesset equation, and (2) presuming that bubbles will develop like an ideal ball.

2.5.2 Schnerr-Sauer cavitation model

The Schnerr-Sauer cavitation model was introduced in 2001 [39]. The beginning point of the Schnerr-Sauer model is vapor volume fraction. The equations of this model are given by equation (36), where $\dot{m}^+$ and $\dot{m}^-$ represent evaporation and condensation during the phase change, respectively.

$$\begin{cases} \dot{m}^+ = \frac{\rho_v \rho_l}{\rho_m} \alpha_v (1 - \alpha_v) \frac{3}{R} \sqrt{\frac{2}{3} \frac{(p - p_v)}{\rho_l}} & (p \geq p_v, \text{condensation}) \\ \dot{m}^- = -\frac{\rho_v \rho_l}{\rho_m} \alpha_v (1 - \alpha_v) \frac{3}{R} \sqrt{\frac{2}{3} \frac{(p_v - p)}{\rho_l}} & (p \leq p_v, \text{evaporation}) \end{cases} \quad (36)$$

Sauer and Schnerr used a simplified Rayleigh-Plesset source terms for the bubble growth and collapse presented in equation (36). The turbulence effect as well as effect of non-condensable gas which have vital importance in diagnosis of inception and growth of cavitation, are considered in this cavitation model.

2.5.3 Singhal cavitation model

The Singhal cavitation model was introduced in 2002 [40]. This model is also known as Full Cavitation model, and it was the first cavitation model to be employed in a commercial software. This model considers a very wide variety of physical parameters such as turbulence kinetic energy, mass fraction of some non-condensable gases such as CO_2 or light hydrocarbons, and surface tension inside its source terms. Furthermore, the beginning point of this model is the mixture density of the fluid. This model has the most simplifications, and the equations of this model are stated as:

$$\begin{cases} \dot{m}^+ = C_p \frac{\sqrt{k}}{\gamma} \rho_l \rho_l \sqrt{\frac{2}{3} \frac{(p - p_v)}{\rho_l}} \cdot \left(\frac{\alpha_v \rho_v}{\rho_m} \right) & (p \geq p_v, \text{condensation}) \\ \dot{m}^- = -C_d \frac{\sqrt{k}}{\gamma} \rho_l \rho_v \sqrt{\frac{2}{3} \frac{(p_v - p)}{\rho_l}} \left(1 - \frac{\alpha_v \rho_v}{\rho_m} \right) & (p \leq p_v, \text{evaporation}) \end{cases} \quad (37)$$

Where, C_p and C_d are the production and destruction coefficients of the Singhal model, and γ is the surface tension coefficient. The constant values in the ANSYS-Fluent solver are assumed to be $C_p = 0.01 \left(\frac{m}{s} \right)$, $C_d = 0.02 \left(\frac{m}{s} \right)$, $\gamma = 0.0717 \left(\frac{N}{m} \right)$, and f_v is the vapor mass fraction $f_v = \frac{\alpha_v}{\rho_m}$.

The Singhal cavitation model could be applied for two-phase cavitation phenomenon modeling only when the mixture multiphase model was selected.

2.5.4 Kubota cavitation model

The last mentioned cavitation model is the Zwart-Gerber-Belamri model which was introduced in 2004 [41]. This model is also called the Kubota model and it is now accessible in the FLUENT solver. The beginning point of the Kubota model is vapor volume fraction. However, it is notable that in the evaporation equation of this model, the product of nucleation site of volume fraction (α_{nuc}) and residual fluid volume ($1 - \alpha_v$) is replaced as an alternative of the vapor volume fraction (α_v). The equations of this model are stated as:

$$\begin{cases} \dot{m}^+ = C_p \frac{3\alpha_v \rho_v}{R} \sqrt{\frac{2(p - p_v)}{3\rho_l}} & (p \geq p_v, \text{condensation}) \\ \dot{m}^- = -C_d \frac{3\alpha_{nuc}(1 - \alpha_v)\rho_v}{R} \sqrt{\frac{2(p_v - p)}{3\rho_l}} & (p \leq p_v, \text{evaporation}) \end{cases} \quad (38)$$

Where, p is local pressure of the fluid, p_v is the saturated vapor pressure, R is the bubble radius, C_p is the constant reconversion rate of vapor to liquid in the zones at which the local pressure exceeds the vapor pressure, and C_d is the constant rate of vapor generation in the zones where the local pressure is less than the vapor pressure. The Kubota cavitation model could be applied when the mixture or full-Eulerian multiphase models are selected. Furthermore, this cavitation model is compatible with all the existing turbulence models. In the current work, the values in the ANSYS-Fluent solver are assumed to be $\alpha_{nuc} = 5 \times 10^{-4}$, $R = 1 \times 10^{-6}$, $C_d = 50$, and $C_p = 0.01$. It is noteworthy that because of the numerical simulation instabilities, vapor volume fraction is replaced by the product of nucleation site volume fraction (α_{nuc}) and remaining fluid volume fraction ($1 - \alpha_v$).

2.5.5 Difference of the mentioned cavitation models

In the case of Schnerr and Sauer cavitation model, the vapor volume fraction is the start point of the source term presented in equation (39).

$$\alpha_v = \frac{v_v}{v_m} \quad (39)$$

However, the beginning point of the Singhal et al. model is analysis of the mixture density and its variation presented in equation (40).

$$\rho_m = \alpha_v \rho_v + (1 - \alpha_v) \rho_l = \alpha_v (\rho_v - \rho_l) + \rho_l \quad (40)$$

On the other hand, the start point of Kubota model is the vapor volume fraction and its change using another formulation presented in equation (41).

$$\frac{d\alpha_v}{dt} = n_0 \cdot 4\pi R^2 \frac{dR}{dt} \quad (41)$$

The Schnerr and Sauer cavitation model is known as the model with the longest analysis time since it contains the largest number of equations [42]. Hence, it is generally known as the most precise model which simulates the cavitation phenomenon transparently. Although, the Singhal cavitation model used the biggest number of simplifications.

Due to the considered simplifications in the Kubota and Singhal model, the empirical constants were employed and their amount change subject to the investigating system and that is why we must calibrate the model if the data for the investigating case is missing.

2.6 Eddy viscosity modification

The standard two-equation turbulence models are capable to model incompressible fluid flow. If we plan to perform our simulation with such a turbulence model, in the regions at which we expect phase change to happen, some modifications in the value of eddy viscosity (μ_t) should be

implemented. To enhance the quality of the numerical simulation of compressible flows using these models, the effect of regional phase change should be considered via the mean density variations. Several non-linear eddy viscosity models have been developed during the past decades. Certain modifications for the $k - \varepsilon$ *RNG* and SST $k - \omega$ turbulence model are presented in this section.

Coutier-Delgosha et al [43] reported that as turbulent dissipation was overvalued in the back of the cavitation sheet, simulation of the cloud cavitation phenomenon with a two-equation turbulence model has some problems. Hence, the reentrant jet blocked prior to breaking the cavity boundary.

In another research, they analyzed the turbulence effect on cavitating flows in a Venturi-type section, Coutier-Delgosha et al [44] proposed a modified versions of the $k - \varepsilon$ *RNG* and $k - \omega$ models with compressibility effects, linked with a barotropic liquid/vapor state law $\rho(P)$ that connects the density variation to the pressure evolution. They proposed to reduce the turbulence viscosity at the interface between the phases and modified the standard $k - \varepsilon$ *RNG* model by diminishing the mixture turbulent viscosity in the minimal void ratio regions through the equation (42).

$$\begin{cases} \mu_t = f(\rho) C_\mu \frac{k^2}{\varepsilon} \\ f(\rho) = \rho_v + \left(\frac{\rho_v - \rho}{\rho_v - \rho_l} \right)^n (\rho_l - \rho_v) \quad n \gg 1 \end{cases} \quad (42)$$

The prediction of the unsteady re-entrant jet with $n = 10$, associated with the modified $f(\rho)$ where the mixture turbulence viscosity was reduced, is presented in Figure 2.3.

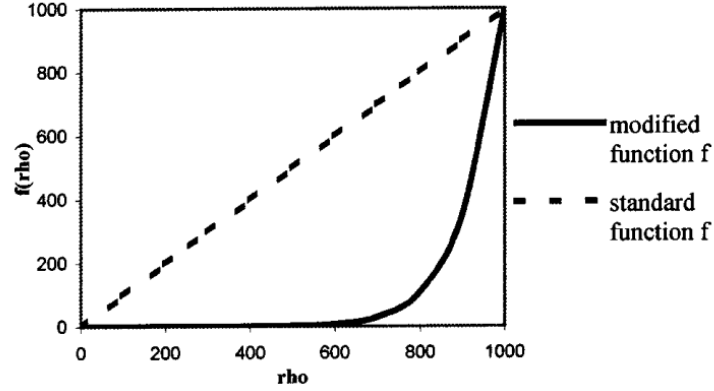


Figure 2.3 The transient function used to modify the local compressibility of the mixture viscosity [44].

Wilcox [30] reported that using the *SST* $k - \omega$ model and the named modification, the unsteady cavitation phenomenon could not be simulated efficiently. He noticed that the Reynolds stress transport equation is concerned with the compressibility effects of the fluid and presented the impact of Mach number in the turbulence equations through equations (43, 44), where, M_t is the turbulence Mach number, A is the local sound celerity, the i index implies the values obtained by the incompressible flow, $M_{t_0} = 0.25$ and $\xi^* = 1.5$.

$$M_t = \sqrt{\frac{2k}{A^2}} \quad (43)$$

$$F(M_t) = \begin{cases} (M_t^2 - M_{t_0}^2) & M_t > M_{t_0} \\ 0 & M_t \leq M_{t_0} \end{cases} \quad (44)$$

On the other hand, Orszag et al. [45] proposed a modification for the $k - \varepsilon$ *RNG* model that could be used in the Eddy viscosity formulation which is:

$$\begin{cases} \mu(t) = f(\rho) C_\mu \frac{k^2}{\varepsilon} \\ f(\rho) = \rho_v + \alpha^n (\rho_l - \rho_v) \end{cases} \quad (45)$$

The equation of $f(\rho)$ defines the density of the two-phase cavitating flow where α is the vapor volume fraction, $n = 10$ and $C_\mu = 0.085$. Then, for the zones, containing only vapor, its value would be equal to ρ_v and for the zones where there is only liquid, its value would be ρ_l .

Ducoin et al. [46] exhibited that the modified *SST* $k - \omega$ with local density correction provided better results in comparison with the original *SST* $k - \omega$ model. The following curvature and turbulent viscosity corrections are employed in the named modified *SST* $k - \omega$ turbulence model. The curvature correction is

$$\begin{cases} f_c = \max\{\min(f_c^*, 1.25), 0\} \\ (1 + c_{r1}) \frac{2r^*}{1 + r^*} [1 - c_{r3} \tan^{-1}(c_{r2} \tilde{r})] - c_{r1} \end{cases} \quad (46)$$

Where, the values of the constants are $c_{r1} = 1.0$, $c_{r2} = 2.0$, and $c_{r3} = 1.0$. The variable quantity of r^* and $\tilde{r}$ are named by Smirnov et al [47].

$$r^* = \frac{S}{\Omega} \quad (47)$$

$$\tilde{r} = 2\Omega_{ij}S_{jk} \left[\frac{DS_{ij}}{Dt} + (\varepsilon_{imn}S_{jn} + \varepsilon_{jmn}S_{in})\Omega_m^{rot} \right] \frac{1}{\Omega D^3} \quad (48)$$

Where,

$$S_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (49)$$

$$\Omega_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right) + 2\varepsilon_{mji}\Omega_m^{rot} \quad (50)$$

$$S^2 = 2S_{ij}S_{ij} \quad (51)$$

$$\Omega^2 = 2\Omega_{ij}\Omega_{ij} \quad (52)$$

$$D^2 = \max(S^2, 0.09\omega^2) \quad (53)$$

Furthermore, the turbulent viscosity correction which is proposed by Courtier-Delgosha et al [43] is:

$$\begin{cases} \mu_{t-mod} = \mu_t f(n) \\ f(n) = \frac{\rho_v + (1 - \alpha_v)^n (\rho_l - \rho_v)}{\rho_v + (1 - \alpha_v)(\rho_l - \rho_v)} \end{cases} \quad (54)$$

Ducoin et al employed the named corrections and considering $n = 3$, proposed the following modified *SST* $k - \omega$ turbulence model.

$$\begin{cases} \frac{\partial}{\partial t}(\rho_m k) + \frac{\partial}{\partial x_i}(\rho_m k u_i) = \frac{\partial}{\partial x_j} \left(\left(\mu_m + \frac{\mu_t}{\sigma_k} f(n) \right) \frac{\partial k}{\partial x_j} \right) + f_c P_k - \rho_m \beta k \omega \\ \frac{\partial}{\partial t}(\rho_m \omega) + \frac{\partial}{\partial x_i}(\rho_m \omega u_i) = \frac{\partial}{\partial x_j} \left(\left(\mu_m + \frac{\mu_t}{\sigma_\omega} f(n) \right) \frac{\partial \omega}{\partial x_j} \right) + \alpha \frac{\omega}{k} f_c P_k - \rho_m \beta \omega^2 + D_{\omega^*} \end{cases} \quad (55)$$

2.7 State of the art

In the past decades many numerical studies were carried out in the domain of unsteady cavitation around hydrofoils. Yet, there are still some gaps to be filled to have numerical predictions with higher precision.

Ji et al. [48] using the LES model coupled with the Schnerr-Sauer homogenous cavitation model performed numerical study of the transient vortical flow around a NACA66 hydrofoil. The numerical predictions of the cavitation shedding behavior from cavitation growth to cavitation-collapses were relatively in good agreement with the experimental results. However, the limitation of their work was that its computational cost was high and that it had excessive computational time.

More specifically, the pressure fluctuation at $x = 0.7c$ was obtained after 6000 time steps unsteady calculations.

In 2015, Tran et al. [8] compared the precision of Kubota and Merkle cavitation models to predict behavior of cavitating fluid around the NACA66 hydrofoil. The Merkle cavitation model which was implemented in the ANSYS-CFX using a UDF code, provided more accurate results in comparison with the only cavitation model available in the ANSYS-CFX commercial solver. Furthermore, Reynolds-Averaged Navier-Stokes (RANS) with standard $k - \varepsilon$ and Shear Stress Transport (SST) models, and Unsteady Reynolds-Averaged Navier-Stokes (URANS) with SAS-SST model, were used to simulate turbulent fluid in steady and unsteady cavitating regimes. The authors reported that the simulation performed with both cavitation models and using the SAS-SST turbulence model shows a more dynamic behavior of cloud cavitation in comparison with the standard $k - \varepsilon$ model and the frequency of the lift fluctuations around the hydrofoil were closer to the experimentally measured amounts in those cases. While the measured peak of the lift fluctuations was reported at around 25% of the cavity-shedding cycle, their simulated peak occurred at around 60% of the cycle. Since the selected RANS turbulence models overpredicts the turbulent viscosity, the authors modified this parameter in another simulation to modify the quality of cavity-shedding predictions.

In 2018, Kadivar et al. [49] performed numerical simulations around the CAV2003 hydrofoil and using the open source OPENFOAM finite volume solver to find the ideal size and location of the cavitation passive controller by which they could make cavitating flow stable or diminish high-pressure amplitudes around the hydrofoil, inhibit the cyclic behavior of unsteady cloud cavitation, and increase the hydrodynamic efficiency of the hydrofoil. In this regard, they used Partially averaged Navier-Stokes (PANS) turbulence model and Schnerr and Sauer cavitation model to numerically predict the behavior of unsteady cloud cavitation without cavitation controller and validated the results with experimental results. Then, they designed six case-studies at which they investigated the mentioned parameters on wedge-type cavitation controllers with different heights (h) and lengths (l) which are placed in various spots (x) on the suction side of the hydrofoil.

The authors reported that in the case study where the values of the defined parameters were ($h=0.0025$, $l=0.011c$, and $x=0.03c$), the dominant frequency of cloud cavity shedding, or pressure

fluctuation frequency was increased by 35% compared to the dominant frequency in the case of regular hydrofoil. However, in the CG-V case study, cavitating flow became stable and no pressure peak was reported on the suction side of the hydrofoil. On the other hand, the frequency of cavitation-induced vibrations was reduced by 95% in comparison with the results obtained using the regular hydrofoil.

Chandel et al [50] studied dynamic stall using URANS and DES models around the NACA0012 airfoil at angle of 10° at a relatively low Reynolds number of $Re=135000$. They investigated the unsteady flow physical characteristics using DES and URANS equations with Menter's *SST $k - \omega$* turbulence model. The numerical results were compared with the available experimental and LES results. They reported that the moment stall and dynamic stalls captured using LES and DES methods were in good agreement with the measured amount. However, the location of dynamic stall predicted using URANS method, was observed earlier than DES, LES, and experiments, respectively. This is while the peak value of the lift coefficient for the DES, LES, and measured values was in a similar range, but the URANS method overpredicted the peak values of the lift coefficient. Furthermore, the maximum value of negative pitching moment was predicted by URANS.

Yu et al. [51] in 2019 numerically studied the effect of Kunz, Zwart, and Full Cavitation Model (FCM) cavitation models on cavitation development around a NACA00015 hydrofoil. They reported that the adverse pressure gradient results obtained by Kunz cavitation model were in good agreement with experimental results and the re-entrant jet was captured accurately in the sheet cavitation. However, they concluded that the Zwart and Kunz cavitation models has a weakness in predicting the unsteady characteristics of the cavitating flow such as cavitation shedding phenomenon. They explained that since the FCM takes turbulent kinetic energy into account to define the condensation source term, it connects the phase transition and turbulent flow field more strongly and that is why this model has strength in predicting the unsteady behavior of the flow.

In another study in 2019, Hidalgo et al. [52] applied Zwart-Gerber-Belamri cavitation model, and the hybrid RANS-LES *SAS - SST $k - \omega$* turbulence model to study the unsteady behavior of cavitation around the NACA66 hydrofoil using the OpenFOAM solver. They validated their numerical predictions with the experimentally measured results and compared them with the results

obtained by the Implicit Large Eddy Simulation (ILES) and the cycle frequency was 4.01 Hz which is reasonably close to the measured value. They reported that using *SAS – SST $k – \omega$* model, the cavitation length, cavity detachment, and frequency of cavitation oscillation were predicted more precisely. Furthermore, according to the pressure coefficient pulsation, the predicted value of vapor volume using *SAS – SST $k – \omega$* was smaller in comparison with the value obtained using ILES model.

In 2019, Li et al. [53] studied flow unsteadiness and kinetic energy transport associated with the cavitation around a two-dimensional NACA66 hydrofoil. They select a modified *SST $k – \omega$* turbulence model and Schnerr-Sauer cavitation to perform the numerical prediction using the OpenFOAM solver. They also used curvature correction and turbulent viscosity corrector which is stated in the section 2.6. Their results revealed that cavitation increased unsteadiness and turbulence intensity around the surface of the hydrofoil in the $\sigma = 1.25$ condition. Furthermore, they illustrated that while the attached sheet cavity was growing, the value of turbulent kinetic energy was reported high at the back end of the cavity region. However, the highest value of turbulent kinetic energy was spotted within the shed cavity.

Some of the important parameters that control cavitation in hydrodynamical structures are surface pressure distribution, nuclei intensity, and initial turbulence level. The other important parameter is surface roughness. Coutier-Delgosha et al [54] in 2005 demonstrated that the surface roughness may alter the cavitation development in the sheet cavitating condition.

In 2010, Li et al. [55] investigated the effect of grooves around a revolving body with experiments and numerical methods. The grooves caused pressure fluctuation at the surface of the body. Based on the numerical results obtained using the ANSYS-Fluent solver, by increasing the groove width, sudden increase in the amplitude of pressure fluctuations was reported around the grooves and the local pressure reduced by increasing the width of groove. In another study in 2016, Churkin et al [56], by visual experiments around a NACA0015 hydrofoil, reported that adjusting surface roughness they could manage to control cavitation development.

In 2015, Kim et al. [57] numerically studied the effect of hydrophobicity on cloud cavitation over the Clark-Y hydrofoil. In this regard, they employed the OpenFOAM solver, and the LES method was used to model turbulence, and the two-phase mixture flow was modeled using Kunz cavitation

model. They concluded that the cloud cavitation became more stable by increasing the hydrophobicity of the hydrofoil's surface. Indeed, their results demonstrated that by increasing the slip strength the drag friction was decreased, the re-entrant jet weakened, the cavity length increased, and the frequency of cavitation shedding decreased.

CHAPTER 3 STEADY-STATE ANALYSIS OF CAVITATING FLOW AROUND THE NACA66 HYDROFOIL

According to complexity of the hydraulic machines, performing numerical simulations to predict the behaviour of cavitation during the different cycles of operation and consequently reducing the maintenance costs is necessary. However, there are still a lot of gaps to be filled so that researchers can make more accurate numerical predictions of this complex phenomenon. In this chapter, we present our proposed methodology to predict the flow characteristics for a steady-state flow regime, performed using the ANSYS FLUENT 16.1 commercial solver. The machine used for performing the numerical simulations was equipped with a 32 GB RAM and Intel(R) Core(TM) i7-4770 CPU 3.40 GHz.

3.1 Presentation of the experimental case study

To evaluate the accuracy of our numerical simulations, we have planned to numerically model cavitating flow around a two-dimensional NACA-66 cambered hydrofoil, located in a computational domain which has the same dimensions as the test section in which Leroux et al [9] performed a similar experimental study of unsteady partial cavitation. The hydrofoil was fabricated of a polished stainless steel and its chord length was $c = 0.15 \text{ (m)}$, the span was $s = 0.191 \text{ (m)}$, and the relative maximum thickness was $\tau = 12\%$. The experiments were conducted in Ecole Navale Cavitation tunnel with a 1m length and 0.192m wide square section (Figure 3.1). It was reported that an inlet velocity up to 15 (m/s) and a pressure in a range of 30 mbar to 3 bars are achievable in the tunnel. The effective angle of attack (α) was 6.5° , and the nominal freestream velocity for the experiments was $U_\infty = 5.33 \text{ (}\frac{m}{s}\text{)}$, in a flow with $Re = 0.8 \times 10^6$.

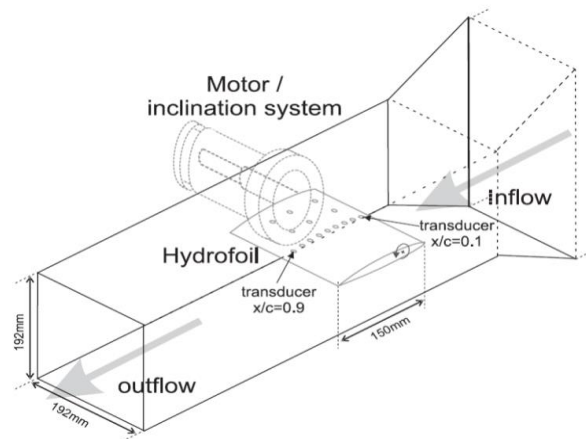


Figure 3.1 The experimental setup revealing the mounted NACA-66 hydrofoil with seventeen piezo-resistive installed transducers in the École Navale cavitation tunnel [9].

They repeated the experiment four times in a condition at which the cavitation number was 1.41, 1.34, 1.30, and 1.25. Based on the obtained results, unsteady behaviour was observed in all the cavitating fluids. They reported the cavity length on these experiments as it is mentioned in Table 3.1. However, the experimental results revealed that in the experiment where the cavity length was more than almost half the chord length, namely $\sigma = 1.25$, the cavity pulsated, its length varied, and the cavity was demonstrated to be unstable. Since the values of experimentally measured pressure coefficient (C_p) in the cavitating flow at $\sigma = 1.25$ was lower than the measured values of C_p at non-cavitating condition all around the hydrofoil surface, the cavity length could not be measured. However, it is reported that the other cavitating flows at $\sigma = 1.41, 1.34$, and 1.30 are stable and later in this chapter, their cavitation length is predicted and reported in Table 3.5.

Furthermore, the peak frequencies, associated with the pressure fluctuations at the cavity closure region, for different cavitation numbers are mentioned in Table 3.1. According to the obtained experimental results, it is reported that the peak frequencies of the pressure fluctuations were reduced as the cavity length increased.

Table 3.1 The measured frequency of the pressure fluctuations intensity on the hydrofoil surface at $x/c = 0.4$ to $x/c = 0.5$, determined using a spectral analysis

σ	l/c	f (Hz)
1.41	0.3	23
1.34	0.4	19
1.30	0.5	14
1.25	> 0.5 (unstable)	-

3.2 Numerical setup

To model the experiments explained above, the following numerical setup is selected and the fluid flow in the considered domain is displayed in Figure 3.2. The data points related to the NACA66 hydrofoil at an angle of attack $\alpha = 6.5^\circ$ are generated using the Javafoil software and they are imported to the ICEM software. A nominal freestream velocity $U_\infty = 5.33 \text{ (m/s)}$ is considered at the inlet. According to the values of density and dynamic viscosity of fresh water at 25° , $\rho = 997.1 \text{ (kg/m}^3\text{)}$ and $\mu = \rho\nu = 0.890 \times 10^{-3} \text{ (Pa.s)}$, the value of the Reynolds number has a value of 8.9×10^6 . Furthermore, the saturated pressure of water vapor in the calculations is considered to be $p_v = 3169 \text{ (Pa)}$. Similar to the computational domain selected in the work of

Tran et al. [58], the total length of the computational domain is $15c$, with an upstream length equal to $5c$.

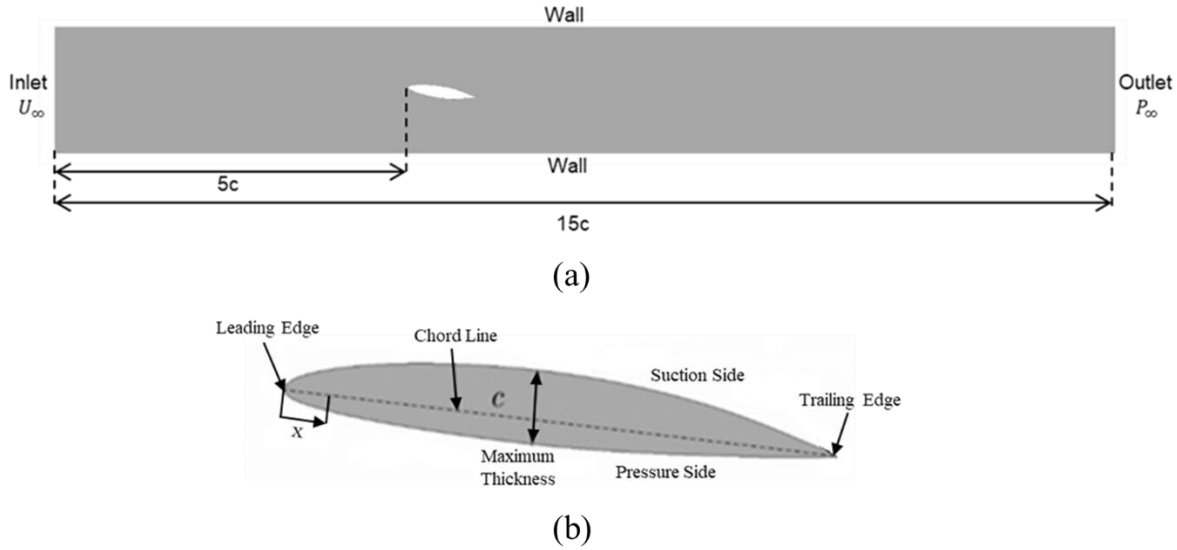


Figure 3.2 (a) Schematic illustration of the numerical test case revealing the boundary conditions and position of the hydrofoil in the designated computational domain, and (b) the details of chosen NACA66 hydrofoil.

A static pressure boundary condition is imposed as the outlet boundary condition, and its value is calculated based on the chosen Thoma number and using $(\sigma = \frac{P_{ref} - P_v}{\frac{1}{2}\rho U_\infty^2})$ equation, and the values are reported in Table 3.2.

Table 3.2 The values of outlet static pressure boundary condition

Thoma Number σ	Outlet Pressure P_∞ (Pa)
1.41	23140
1.34	22148
1.30	21581
1.25	20873

Furthermore, the no-slip wall boundary conditions are imposed on the surface of the hydrofoil and the bottom and top boundaries of the test section. Yet again, according to the measured amount of turbulent viscosity in the experiments, in all the simulations the turbulent intensity in the current research is considered to be 2%. Furthermore, the coupled implicit solver was selected to solve the momentum and pressure equations to promote a quick convergence as well as performing a robust computation. Then, the QUICK discretization scheme in space, PRESTO scheme was used to compute pressure on the cell-faces, and first-order implicit schemes were chosen to obtain reliable predictions.

3.3 Generation of Computational grid

The hexa-structured meshes were generated using the ICEM mesher in the defined computational domain with high refinement around the hydrofoil (Figure 3.3). It is noteworthy that based on the selected $k - \varepsilon$ turbulence model, to use a wall-function near the hydrofoil, a value of first cell spacing is set based on the $Y^+ = 30$. On the other hand, to perform another set of simulations using the $SST\ k - \omega$ turbulence model, a mesh was generated based on a $Y^+ = 1.00$.

Using equation (56), the values of the first cell distance (y) is calculated to be 0.0001245 (m), and 0.00000415 (m), based on a Y^+ value of 30 and 1.00, respectively.

$$Y^+ = \frac{yu_\tau}{\nu} \quad (56)$$

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} \quad (57)$$

Where, u_τ is the wall friction velocity, ν is the kinematic viscosity, τ_w is wall shear stress and its value is calculated using $(\tau_w = \mu \frac{du}{dy})$ equation.

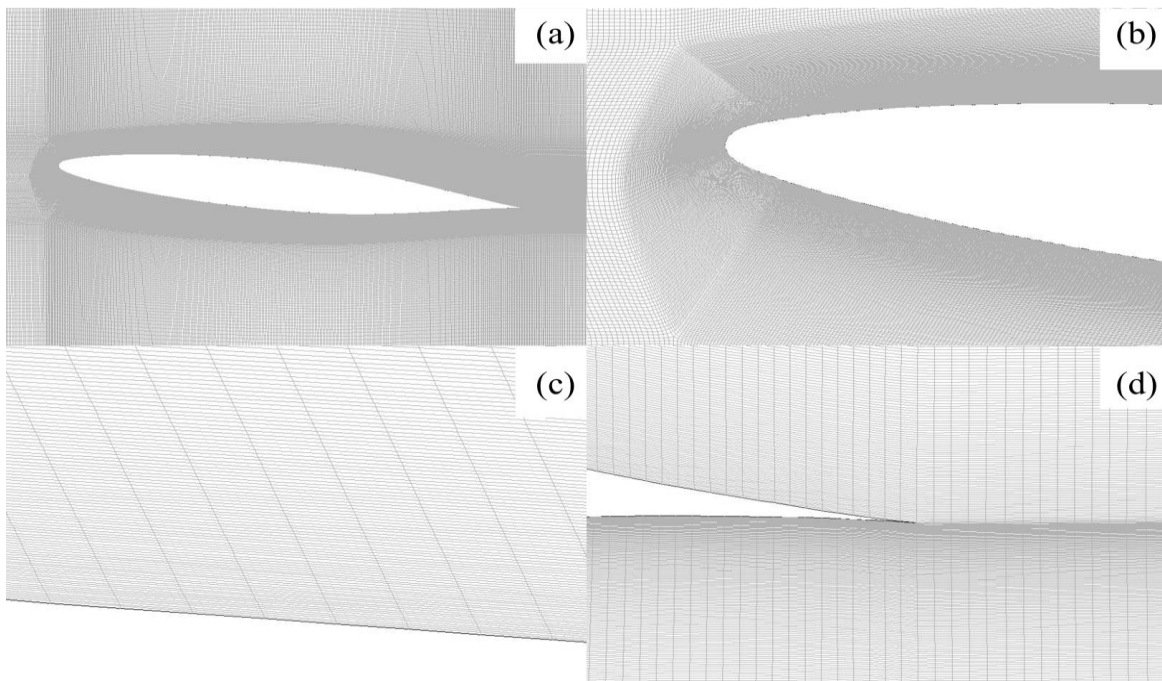


Figure 3.3 Images revealing the details of the computational grid in (a) an overall view, (b) around the leading edge, (c) in the boundary layer at about half chord length, and (d) around the trailing edge of the NACA66 hydrofoil with the angle of attack of 6.5° , for the fine mesh associated with $Y^+ = 30$.

Three different meshes were generated to measure the impact of computational grid resolution on the precision of our calculations (Table 3.3). Initially, a coarse mesh was generated to perform the numerical simulations. Then, the computational mesh was refined in all directions using a $\sqrt{2}$ factor and a medium mesh was generated. Finally, using a refinement factor of 2 in every direction, the fine mesh was created.

Table 3.3 Details of the numerical grids for the hydrofoil section in the generated computational domain

	Coarse mesh	Medium mesh	Fine mesh
Number of nodes	252,018	503,329	1,005,244
Number of quads. cells	249,887	501,440	1,002,572

3.3.1 Mesh quality

Given the significant impact of computational grid on the accuracy of the solution, a mesh study is of great importance. In this regard, the mesh criteria should be within the proposed range. The mesh metrics available in the ICEM CFD mesher that are investigated in the current research are Aspect Ratio (AR), Orthogonal Quality (OQ), smoothness, and skewness. It should be noted that the skewed meshes are those which have a greater aspect ratio. These criteria are introduced and explained in detail in Appendix A. The introduced mesh qualities for all the meshes employed in the current research are presented in Table 3.4.

Table 3.4 Investigation of the fine mesh qualities based on the mesh metrics offered by ANSYS

	Reported Value	Acceptable Range	Condition
Aspect Ratio	130	$\sim 1 (< 1000)$	ok
Orthogonal Quality	0.38	$0.15 < OQ < 1$	Good
Smoothness	1.24	$1 < \sigma < 2.5$	Good
Skewness	0.61	$0 < S < 0.94$	Good

3.4 Principal case-studies in the current work

In the first step, to assure the precision of numerical simulations using both steady-state and transient algorithms, and to cope with the complexity of the cavitation phenomenon, the numerical simulations for a non-cavitating condition were performed and validated with the experimental results.

Then, another set of simulations were performed in different case studies with the selected cavitation model. They were calculated with Thoma numbers for which the experimental results are reported, namely $\sigma = 1.41, 1.34, 1.30$, and 1.25 , and the results were compared with the results, reported by Leroux et al. To this end, the numerically calculated pressure coefficients should be compared with the test values.

Finally, to track the formation and collapse behaviour of the vapor bubbles in different time-steps during the computation, the transient algorithm was implemented in all the mentioned case studies.

In the following section, some light is shed on the earlier mentioned case studies performed using steady-state algorithm. The results obtained by transient algorithm are presented and discussed in the chapter 4.

3.4.1 Steady-state analysis of fluid flow at non-cavitating condition

To check the accuracy of the chosen turbulence model, the phase interaction was neglected in the first step. The *standard* $k - \varepsilon$ (SKE) turbulence model, which is a robust, commonly used, and reasonably accurate model, was used to model the turbulence in the fluid flow. It is noteworthy that this simulation converged in around 3000 iterations and the simulation was computed in about two hours.

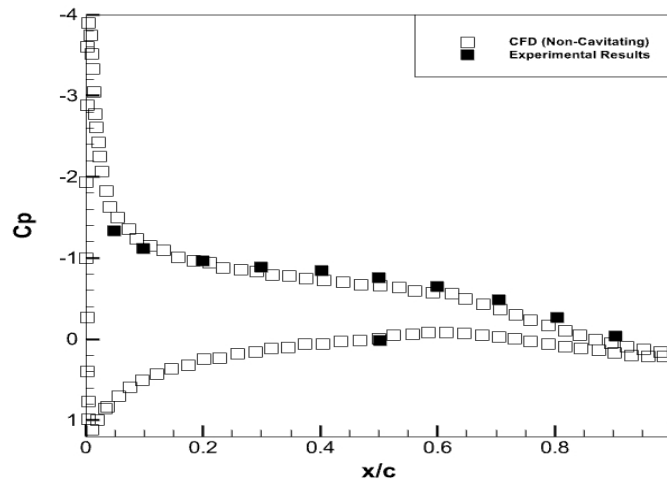


Figure 3.4 Comparison of predicted surface static pressure coefficient values in non-cavitating condition against the experimental results from Leroux et al [9].

Figure 3.4 shows the time-averaged pressure coefficient (C_p) around the surface of the hydrofoil in which the x-axis represents the location on the hydrofoil, and the y-axis represents the value of pressure coefficient. The $x/c = 0$ and $x/c = 1$ are the location of the leading edge and trailing edge of the hydrofoil, respectively. The experimental results were measured using pressure transducers, placed at mid-span most of them on the suction side. Furthermore, the isolated point in the experimental results relates to the transducer on the pressure side of the hydrofoil. As illustrated in the Figure 3.4, the numerical results obtained in the non-cavitating condition, resulted in an excellent agreement with the experimental results.

3.4.2 Steady-state analyzes of fluid flow at cavitating condition

Beside the numerical parameters, discussed earlier on this chapter, there are three important characteristics, namely the mesh density, convergence criteria and turbulence model, that are studied, and the results are presented in the next three sections.

In this section, a steady-state analysis of cavitating flow with different Thoma numbers is reported. It should be noted that the fine mesh was employed to discretize the geometry. Furthermore, according to the good convergence occurred using the standard $k - \varepsilon$ model, Kubota cavitation

model as well as convergence criteria of 1×10^{-5} in terms of continuity, x -velocity, y -velocity, k , ε , and v_f (vapor), were selected and used to perform all the simulations presented in this section.

In Figure 3.5, there are slight differences between the numerically predicted and measured pressure coefficient values. For the cavitating flow at the cavitation number $\sigma = 1.41$, our numerical predictions fall within seven error bars out of the ten pressure points measured on the suction side of the hydrofoil. It should be noted that the pressure coefficient was over-predicted at $x/c \sim 0.3, 0.4$, and 0.7 in the closure region of cavitation. Furthermore, pressure coefficients in the cases of cavitating flow at the cavitation numbers of $\sigma = 1.34, 1.30$, and 1.25 fall in the error bars in eight, six, and four error bars, respectively. According to the obtained results, uncertainties mostly occurred in the cavitation closure region. Moreover, as the cavitation number decreased the accuracy decreased and in the case of cavitating flow at the cavitation number $\sigma = 1.25$, related to the unstable cavitation we observed that the results were not reliable.

As a rule of thumb, steady-state simulations under the stable cavitation condition, generally predict the pressure coefficient well. Although, in the unstable cavitation test case, namely at $\sigma = 1.25$, there exist greater difference between the numerical predictions and the measured values of pressure coefficient. The reason for the lower precision on this case study, specially at the range of $0.1 < x/c < 0.6$ is probably related to the low capability of SKE turbulence model in predicting eddy viscosity in the cavitation closure region on the suction side of the hydrofoil.

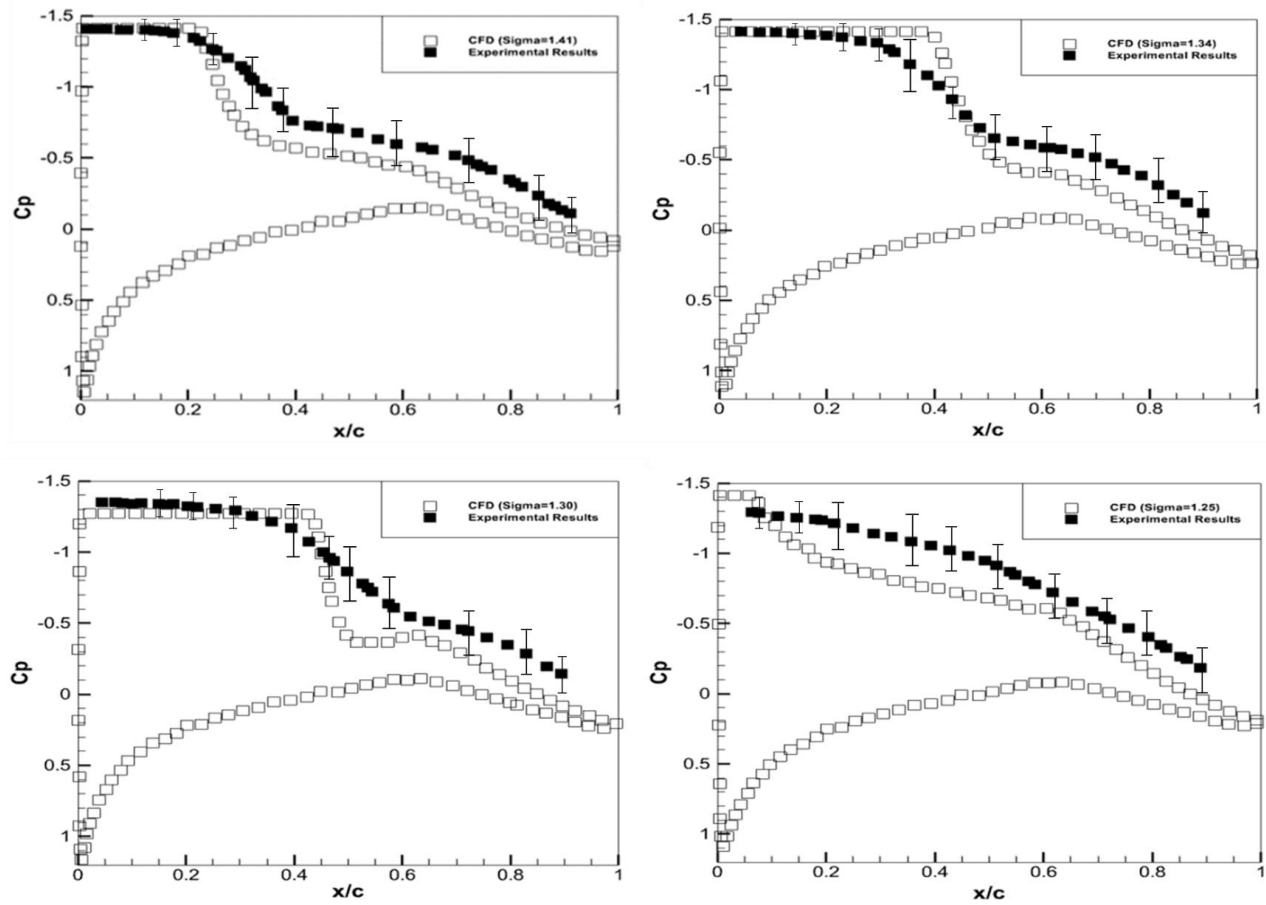


Figure 3.5 Validation of the predicted surface static pressure coefficient distribution in various cavitating flows at different cavitation numbers with the experimental results.

According to the non-cavitating results, which were in excellent agreement with the experiments, it could be concluded that the reason of this uncertainty could also be related to the mass transfer term in the cavitation model, particularly the shedding of cavity bubbles after they collapsed. Furthermore, in the steady-state simulation of cavitating flow at $\sigma = 1.30$ and 1.25 , it is observed in our simulation that the magnitude of the pressure coefficient increased suddenly with a higher slope than the measured values at $x/c \cong 0.4$ and $x/c \cong 0.1$, respectively. The difference between the numerical and experimental results could be reduced by modifying curvature and eddy-viscosity.

To elucidate the behavior of cavitating flow with $\sigma = 1.41$, the velocity vectors are drawn around the NACA66 hydrofoil and some images are presented in the Figure 3.6. In the regions of Figure 3.6(a) where flow has the maximum speed, the pressure has the minimum value, and the cavity bubbles are going to be initiated. Furthermore, it is shown in the Figure 3.6(b) and Figure 3.6(c) that flow reversal appears near the leading edge at around $0.3c$ and it propagates towards the trailing edge. This flow is caused by the re-entrant jet, and this phenomenon is responsible for the unsteady cyclic behavior of the cavitating flows.

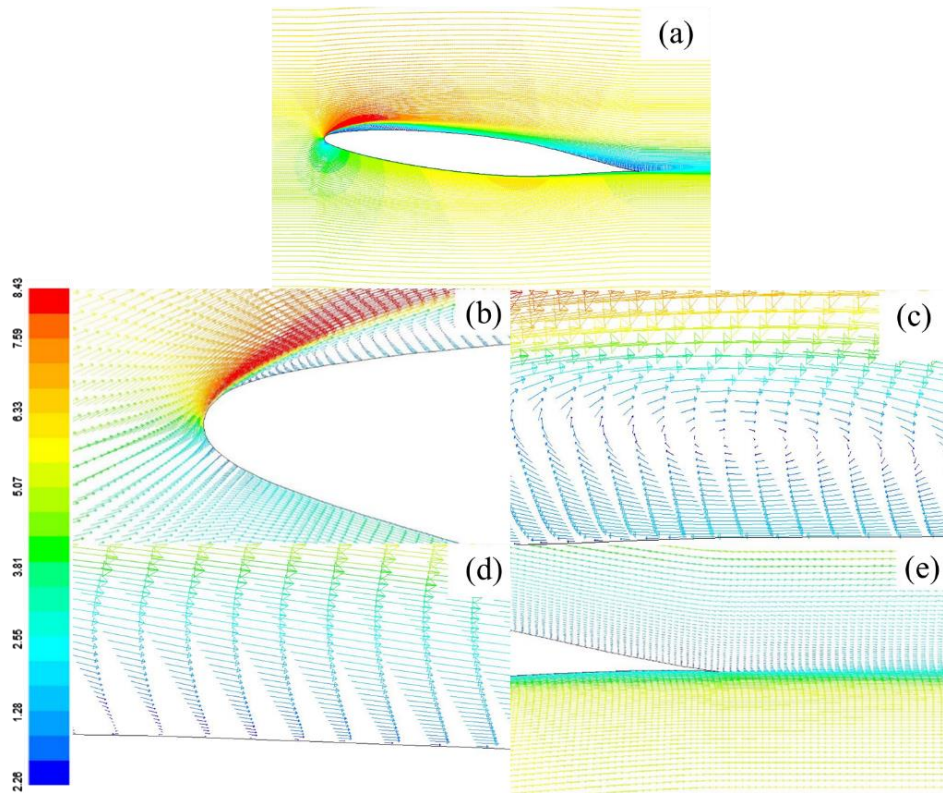


Figure 3.6 Illustration of the flow domain in (a) an overall view around the hydrofoil, (b) vicinity of the leading edge, (c) the suction side, (d) the boundary layer on the suction side, and (e) around the trailing edge, predicted using the SKE turbulence model, Kubota cavitation model and at the $\sigma = 1.41$.

In the Figure 3.6(c), it is shown that the reverse flow generated due to creation of re-entrant jet and during the cavity break-off. Last but not the least, in the Figure 3.6(d), the boundary layer is captured at around half chord length, and it shows that on this length, the reverse flow was halted, and flow has attached boundary layer after this region.

3.4.2.1 Comparison of the obtained numerical predictions

In order to evaluate the difference between the performance of different solvers, the predictions made by our numerical method are compared with those predicted using ANSYS CFX by Tran et al [58]. It is noteworthy that the SKE turbulence model and Kubota cavitation model is selected to perform steady-state simulations and the simulations are performed to simulate the cavitation behavior around a hydrofoil with the same details in case of cavitating flow at $\sigma = 1.49$.

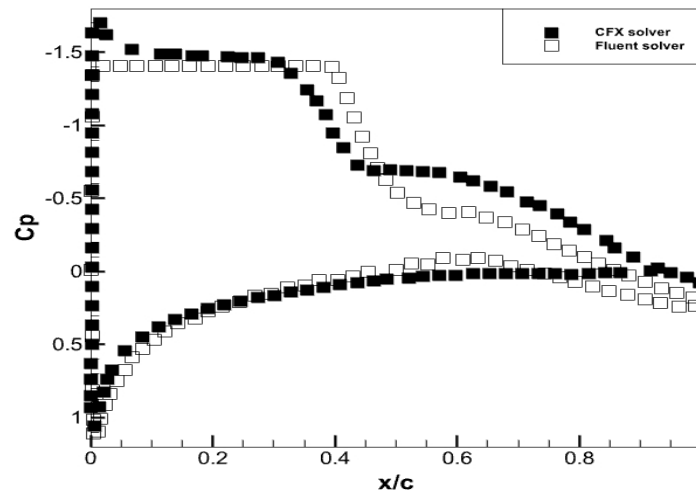


Figure 3.7 Numerical predictions for surface pressure coefficient distribution in a cavitating flow at $\sigma = 1.49$ to compare the results obtained by different solvers

According to the results showed in Figure 3.7, there are some differences specially in the range of $0.35 < x/c < 0.55$ in the cavitation closure region between the predictions which are probably related to the different computational mesh that is used in two different simulations.

3.4.2.2 Impact of computational grid on the numerical predictions

To perform the mesh independency test, the cavitating flow at $\sigma=1.41$ is intended and the steady-state simulations with identical parameters were repeated using all three fully structured computational meshes and the results are reported in Figure 3.8. It should be noted that the standard $k - \varepsilon$ turbulence model and Kubota cavitation model were employed to perform these simulations. To this end, the mesh was generated based on $Y^+ = 30$ to work with a wall function.

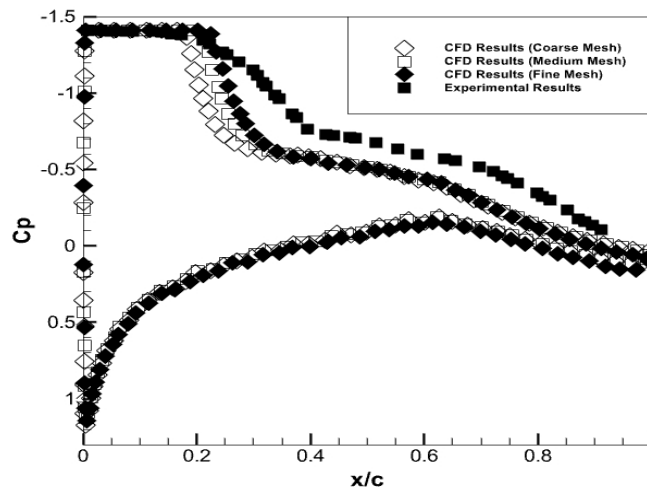


Figure 3.8 Numerical predictions for surface pressure coefficient distribution in a cavitating flow at $\sigma = 1.41$ performed with structured computational mesh with various densities, using standard $k - \varepsilon$ turbulence model and Kubota cavitation model.

Based on the results in Figure 3.8, the solution is only slightly dependant on the computational mesh and almost identical results were obtained for all three meshes.

3.4.2.3 Convergence criteria and their impact on precision of the numerical results

The other important parameter in the current numerical setup is convergence criteria. In the current study, using standard $k - \varepsilon$ turbulence model and Kubota cavitation model with $\sigma=1.41$, the effect of convergence criteria is investigated in four case studies, namely convergence criteria in terms of continuity, x-velocity, y-velocity, k, ε , and v_f (vapor), 1×10^{-3} , 1×10^{-4} , 1×10^{-5} , and 1×10^{-6} .

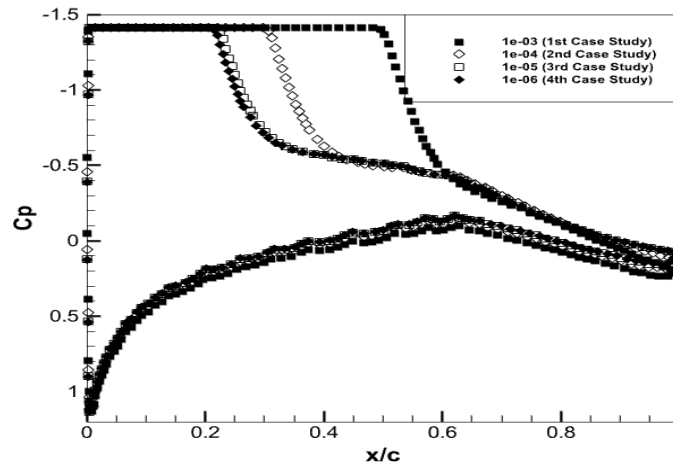


Figure 3.9 Study of the effect of convergence criteria on accuracy of predictions using the structured fine mesh, standard $k-\epsilon$ turbulence model and Kubota cavitation model at $\sigma=1.41$

Figure 3.9 shows that there is a significant difference between the predictions made by the first and forth case study, namely 1×10^{-3} and 1×10^{-6} . Although, the numerical predictions of pressure coefficient are almost identical for the third and forth case study, related to the convergence criteria of 1×10^{-5} and 1×10^{-6} . Hence, since the simulations were performed with almost the same precision and they were less time-consuming, using convergence criteria of 1×10^{-5} , this convergence criteria are selected in the current study.

3.4.2.4 Investigation of the impact of turbulence models

The numerical simulations are performed using two turbulence models, namely standard $k - \epsilon$ and Shear Stress Transport (*SST*) $k - \omega$ models. Then, a comparison is made between the pressure-coefficient obtained using different turbulence models and the ones measured using transducers, presented in the Figure 3.10. The predictions made using standard $k - \epsilon$ model is closer to the experimental results. According to the conclusion is made in the current section, the standard $k - \epsilon$ model is selected in the current research.

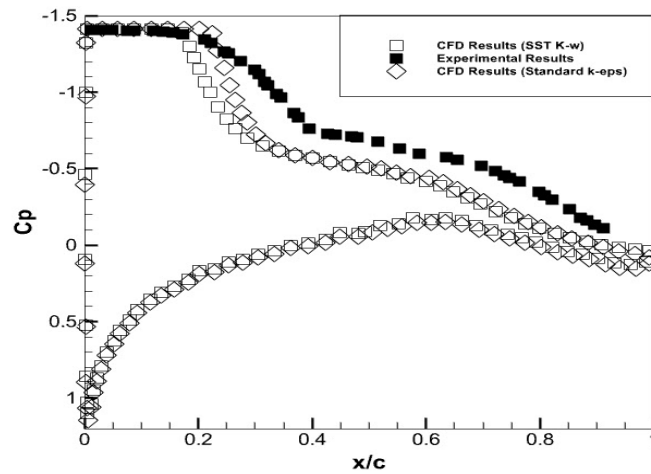


Figure 3.10 Validation of the numerical predictions presented using standard $k - \varepsilon$ and $SST k - \omega$ turbulence model and Kubota cavitation model with $\sigma = 1.41$, against experiments

Although both turbulence models demonstrated good results which agreed with the experimental results, since the SKE turbulence model is well known for its good performance in the industrial applications and considering that the provided results with this model are closer to the measured ones, in comparison with the predictions presented using $SST k - \omega$ model, this model is recognized as a better choice for performing simulation in the current research.

3.4.2.5 Impact of Thoma number on cavitation length

The cavitation length could be predicted using Figure 3.11 in which the numerical prediction related to both cavitating and non-cavitating conditions are reported. As shown in Figure 3.11, the cavitation length (l) is approximately equal to the portion of the hydrofoil surface on the suction side at which the magnitude of the pressure coefficient is greater than the same value in the non-cavitating condition. It is noteworthy that since the cavitation at $\sigma = 1.25$ was unstable, the diagram is not reported for this Thoma number.

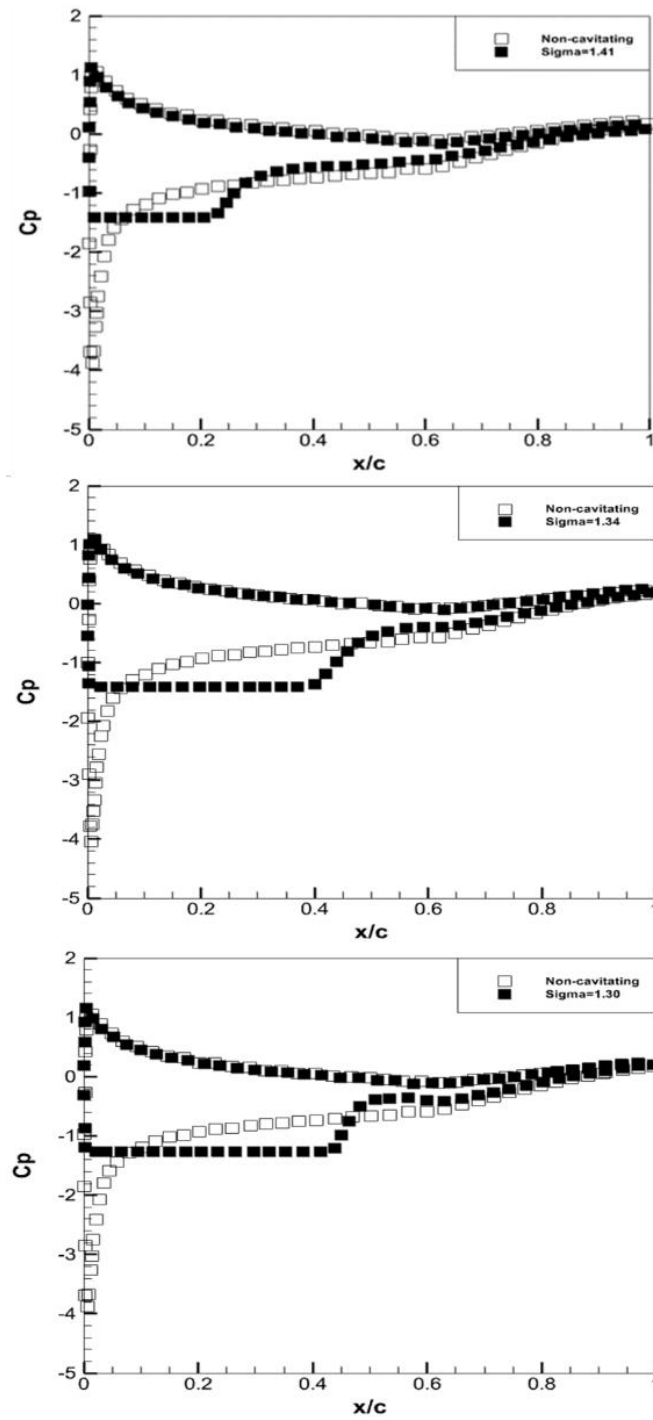


Figure 3.11 Predicted surface static pressure coefficient distribution for the cavitation versus non-cavitating condition at different Thoma numbers

Based on the methodology mentioned earlier in this section and the numerical results presented in Figure 3.11, an estimation of the cavity length for all the stable cavities are reported in the Table 3.5.

Table 3.5 The measured and predicted values of cavitation length in the cavitating fluid flows

		$\sigma = 1.41$	$\sigma = 1.34$	$\sigma = 1.30$
Cavitation length (l/c)	Measured	0.3	0.4	0.5
	Predicted	0.26	0.42	0.43
	Error	13.34%	5%	14%

According to Table 3.5, the best prediction was made for the cavitation length at $\sigma = 1.34$ which is only 5% lower than the measured value. Although it is generally expected that with the same numerical setup, the results associated with a higher cavitation number should be predicted with more accuracy. However, the other cavity-lengths were fairly well predicted. The reason could be related to various contributing factors such as the effect of eddy viscosity and curvature on the precision of cavitation modeling. To elaborate more on it, since the viscosity is overestimated in the boundary layer, it is likely that due to the complexity of the cavitation phenomenon, this viscosity over-prediction behaviour does not show similar behaviour at certain cavitation number.

To shed some light on the effect of Thoma number on the fluid behaviour, the pressure and vapor volume fraction contours related to two cavitation numbers, namely $\sigma = 1.41, 1.25$, namely the smallest and the largest cavitation numbers, respectively, are presented in Figure 3.12.

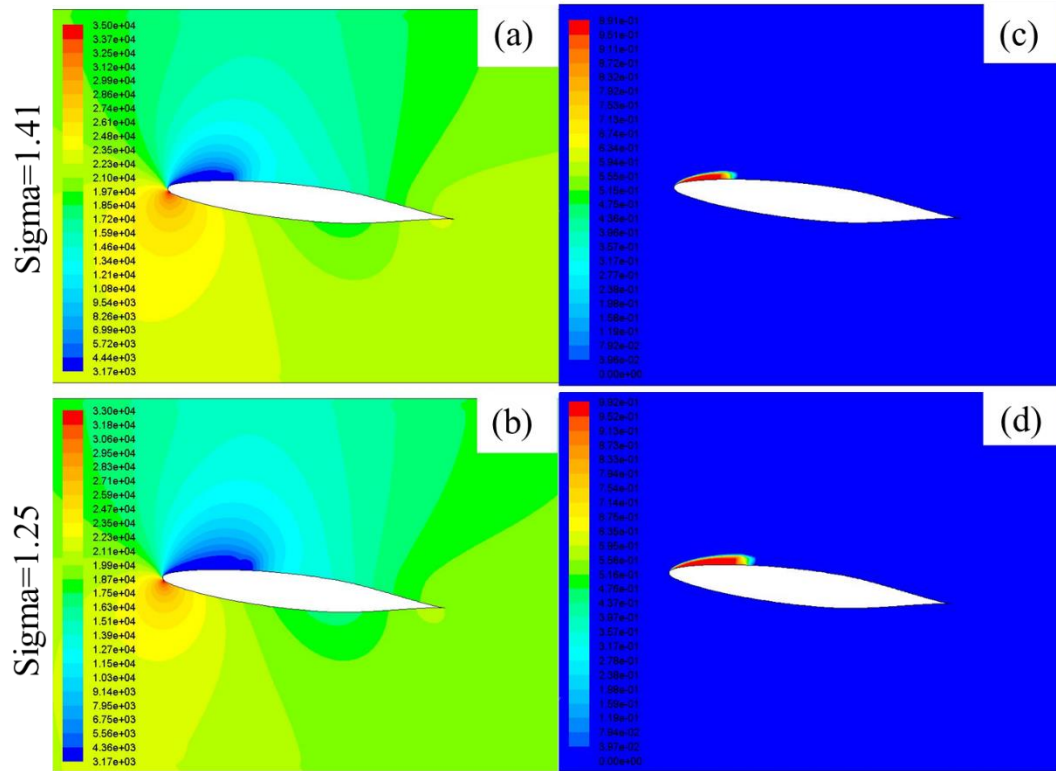


Figure 3.12 Steady pressure contours for cavitating flow at (a) $\sigma = 1.41$, and (b) $\sigma = 1.25$, and vapor volume fraction contours for (c) $\sigma = 1.41$, and (d) $\sigma = 1.25$, around the hydrofoil

As it is illustrated in Figure 3.12(a) and Figure 3.12(b), by decreasing the cavitation number, a wider region near the leading edge demonstrated to have low pressure which confirms the experimental observations reported in Table 3.5. Figure 3.12(c) and Figure 3.12(d) show that the vapor fraction is increased by reducing the cavitation number.

CHAPTER 4 TRANSIENT ANALYSIS OF CAVITATION AROUND THE NACA66 HYDROFOIL

Since turbulence is always unsteady, steady-state simulations in Chapter 3 only provided statistically averaged variables. On the other hand, to study the time history of the water flow over the hydrofoil, transient simulations are performed and presented in the current chapter. In the simulations with the transient algorithm, the ultimate state of the flow is affected by its initial state. In this regard, it is noteworthy that the transient simulations are performed after accomplishing a steady-state simulation for each cavitating flow.

It is also noteworthy that according to the conclusions made in the chapter 3 regarding higher accuracy of the results obtained using a fine computational mesh, all unsteady simulations in the current chapter are calculated using a fine grid.

Furthermore, Schnerr-Sauer and Kubota cavitation models were tested to perform the numerical simulation. However, according to the fact that the convergence occurred only when the Kubota model was employed, this model was used to predict cavitation behaviour for different time-steps. As for the turbulence model, several models such as the Standard $k - \varepsilon$ (SKE), the *SST* $k - \omega$, the Reynolds Stress Model (RSM), and the Scale – Adaptive Simulation (SAS – SST) models were tested. But since the best convergence for the transient simulations was occurred using the SKE turbulence model, this model was used in all simulations.

4.1 Time step study

One of the most important parameters to precisely model transient simulation is to find an adequate value for the time-step size. In the current study, we have modeled fluid flow over a NACA66 hydrofoil during two seconds. In this section, three test cases for the time-step size namely 0.002 (s), 0.0002 (s) and 0.00002 (s) are presented to study the behaviour of the fluid flow throughout the simulation time (Table 4.1).

Table 4.1 Details of the case studies to perform the time step study using the fine mesh

	1 st Case Study	2 nd Case Study	3 rd Case Study
Time step size	0.002	0.0002	0.00002
Number of time steps	1000	10000	100000

Then, in the following two sections, the effect of time-step size is evaluated in non-cavitating and cavitating regimes, respectively. Similar to steady-state simulations, the CFD results are validated with experimental results provided by Leroux et al [9].

4.2 Transient analysis for the non-cavitating flow

Every cavitating simulation is first started with a fully wetted or non-cavitating simulation, and the cavitating flows are simulated following gradual reduction of outlet pressure to the specific values for each cavitating condition. The numerical predictions for the unsteady non cavitating condition is presented in Figure 4.1.

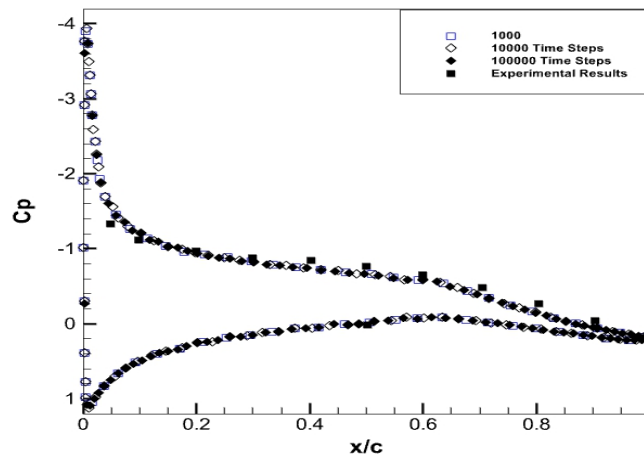


Figure 4.1 Time-averaged static pressure coefficient distribution with SKE turbulence model in three case studies with various time-step sizes for a non-cavitating flow

4.3 Transient analysis for cavitating flows

In section 4.2, it was observed that the predicted static pressure coefficient distribution was virtually independent of time-step size for non-cavitating flow. Since However, this is not the case for cavitating flow as shown in Figure 4.2. It should be noted that the predictions presented in the Figure 4.1 are obtained at the time 0.24 (s), and similar to the experimental study performed by Leroux et al [9], a cavitating flow associated with $\sigma = 1.34$ was chosen to perform the transient simulations, presented in this section. The reason of this selection is related to the fact that the length of cavities in the flow associated to $\sigma = 1.34$ is longer when compared to the cavity length associated with $\sigma = 1.41$.

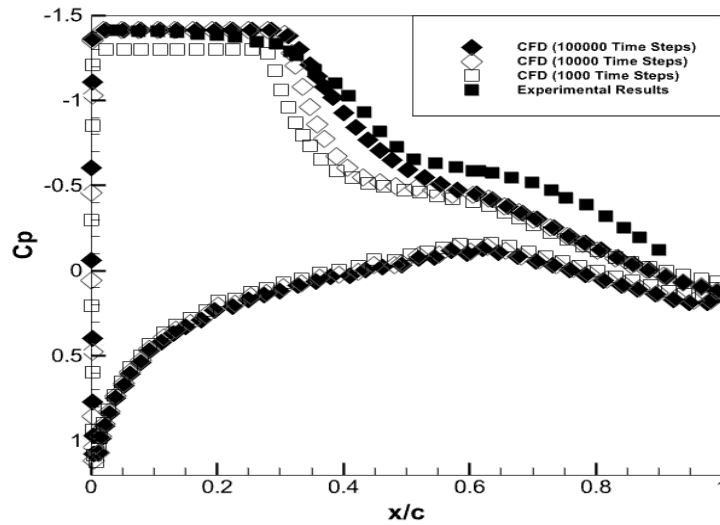


Figure 4.2 Time-averaged pressure coefficient for a cavitating flow at $\sigma = 1.34$ with SKE turbulence model in three case studies with various time-step sizes

Based on the CFD predictions presented in Figure 4.2, the current transient simulation specially in the range of $0.25 < x/c < 0.6$ has a dependency on the time-step size, and it reveals that however the obtained results are reliable, there are some uncertainties associated with them. It is noteworthy that the transient simulation with 1000000 time-steps and time-step size of 0.000002 diverged. Based on these results, the best time step selection corresponds to 100000 time steps.

In section 4.4, the frequency of pressure fluctuation intensity is investigated and the predicted frequencies using different time step sizes are compared and reported, which reveals the crucial impact of choosing an appropriate time-step size on the precision of transient numerical simulation.

4.3.1 Illustration of the cavity-shedding phenomenon

In order to qualitatively study vapor fraction (α_v) for various time-steps and to illustrate the different stages of cavitation phenomenon namely formation, development, shedding, and collapse of cavity bubbles during a cavity-shedding cycle (T), the Vapor Volume Fraction (VVF) contours are reported at several time steps in Figure 4.3. Where, the time difference between each two time step is $T/9$.

Based on the experimental observations, attached cavities were shed from t_a to t_d , and attached sheet cavities develop in close proximity to the leading edge and grows to the trailing edge. Then, attached cavity grows from time-step t_e to t_g , and covers all over the suction side of the hydrofoil, and the re-entrant jet, developed in the closure region, triggers cloud cavity to roll and shed-off. Finally, collapse of cloud cavities occurs from t_h to t_i .

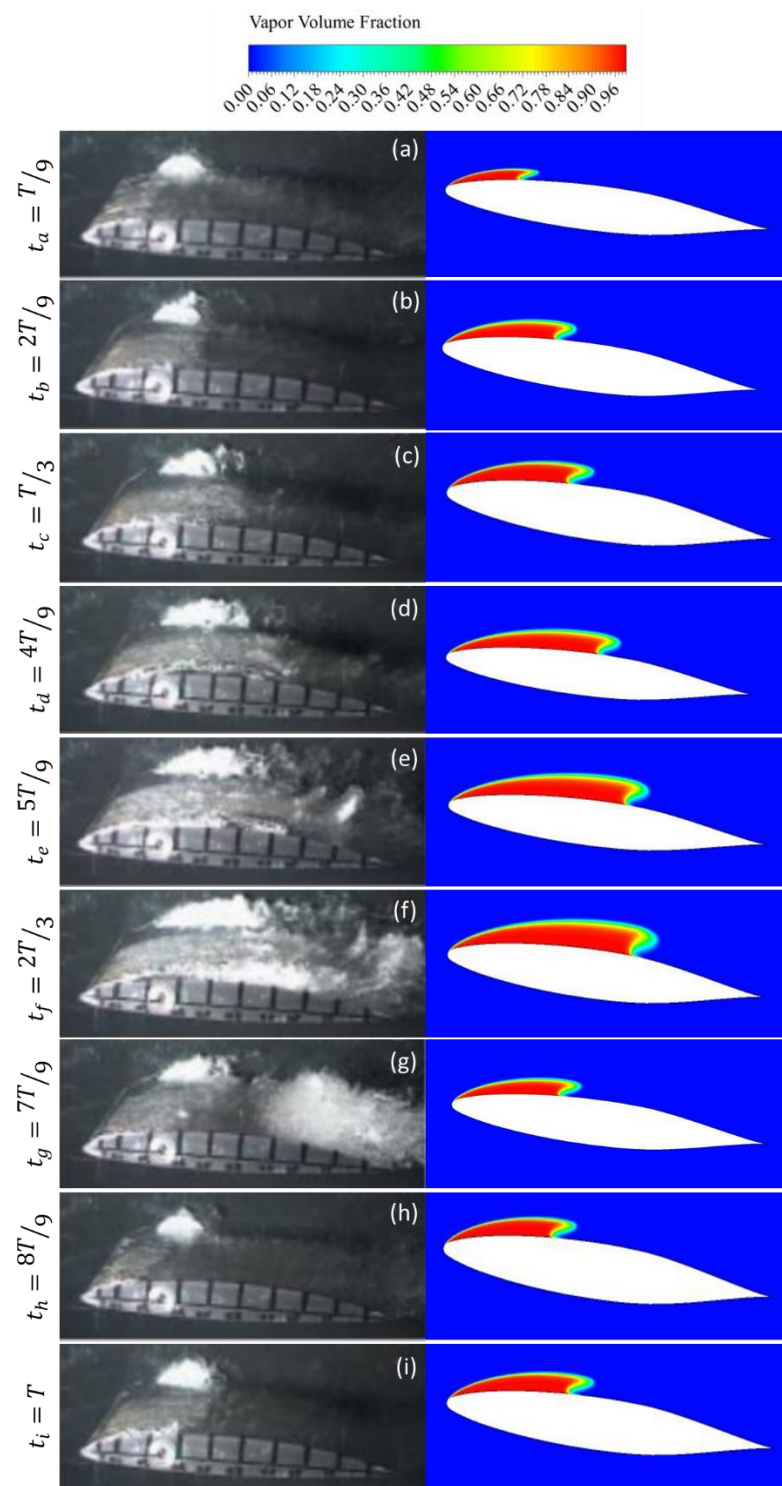


Figure 4.3 Comparison of the Vapor Volume Fraction (VVF) contours, obtained using CFD numerical predictions with the experimentally cavitation evolution in a cavitation cycle

In accordance with the theory, at the end of a cavitation cycle, it is indicated in Figure 4.3 that the total vapor volume is reduced, and it indicates that this model predicted fairly well the bubble collapse. All in all, the current numerical simulation already predicts formation, development, and collapse of cavity bubbles.

Although, given the employed SKE turbulence model and considering the limitations of RANS-based turbulence models, the cut-off cavities are attached to the suction side of the hydrofoil and the simulation was not capable of demonstrating the cavity-shedding phenomenon. Another issue is that the cavitation initiation was predicted by analyzing the pressure coefficient (C_p) diagram over the hydrofoil's chord length, but in Figure 4.3, it is revealed that the attached cavity initiates from the leading edge and it does not show the exact location where cavities primarily occur.

4.4 Study the frequency of sudden pressure fluctuations

Another useful and accurate method to indicate the locations at which the cavitation is likely to occur, is studying frequency of sudden pressure fluctuations. These frequencies were experimentally measured with sensors, and they were studied using spectral analysis [9]. The spectral analysis for the cavity length of 0.4 revealed that the frequency of pressure fluctuations intensity rises at the cavity and closure was reported to be 19 (Hz) at $x/c = 0.4$, and this value remained almost the same until $x/c = 0.5$. However, Leroux et al reported that the frequency was spread at $x/c = 0.6$, and finally no peak frequency was detected at $x/c = 0.7$. Hence, three monitor points were positioned at $x = 0.4c$ on the hydrofoil suction side and the numerical data was collected. Then, Power Spectral Density (PSD) analyses, where the x-axis represents frequency (Hz), and the y-axis represents $10 \log \left[\frac{S_{pp}(f)}{(\frac{1}{2}\rho U^2)^2} \right]$ (dB), are performed to predict peak pressure fluctuations in different cavitating conditions.

Although according to the Figure 4.3, the present numerical setup has some limitations in prediction of cavity shedding, according to the PSD results, it has good accuracy in predicting cavity behaviour at the $x = 0.4c$ and we expect to get CFD results close to the measured data.

According to the results, obtained using PSD analyses, the frequencies associated with the peak frequency fluctuations for the selected cavitation numbers, namely $\sigma = 1.41, 1.34, 1.30$, and 1.25 , are reported in the following tables.

Table 4.2 The measured and predicted values of the pressure fluctuation intensity at the cavity closure of hydrofoil ($x/c = 0.4$), for the cavity length of $l/c = 0.3$ when $\sigma = 1.41$, obtained using a fine mesh

	Experimental Result	CFD Result Time-Step Size 0.002	CFD Result Time-Step Size 0.0002	CFD Result Time-Step Size 0.00002
Frequency (Hz)	23	25.7	24.5	23.8
Error	~3%	11.7%	6.5%	3.5%

Table 4.3 The measured and predicted values of the pressure fluctuation intensity at the cavity closure of hydrofoil ($x/c = 0.4$), for the cavity length of $l/c = 0.4$ when $\sigma = 1.34$, obtained using a fine mesh

	Experimental Result	CFD Result Time-Step Size 0.002	CFD Result Time-Step Size 0.0002	CFD Result Time-Step Size 0.00002
Frequency (Hz)	19	22.9	21.1	20.2
Error	~3%	20.5%	11%	6.3%

Table 4.4 The measured and predicted values of the pressure fluctuation intensity at the cavity closure of hydrofoil ($x/c = 0.4$), for the cavity length of $l/c = 0.5$ when $\sigma = 1.30$, obtained using a fine mesh

	Experimental Result	CFD Result Time-Step Size 0.002	CFD Result Time-Step Size 0.0002	CFD Result Time-Step Size 0.00002
Frequency (Hz)	14	17.4	16.3	15.4
Error	~3%	24.3%	16.4%	10%

It should be noted that since the cavitation shows unstable behaviour in the cases at which cavity length is larger than $0.5c$, including the cavitating flow at $\sigma = 1.25$, the experimental results were not measured. However, to compare pressure fluctuation behaviour of the flow at $x=0.4c$ in an unstable condition, the numerical simulations are performed in three test cases with different time-step sizes and the results are presented in Table 4.5.

Table 4.5 The measured and predicted values of the pressure fluctuation intensity at the cavity closure of hydrofoil ($x/c = 0.4$), for the cavity length of $l/c > 0.5$ when $\sigma = 1.25$, representing an unstable cavitation regime, obtained using a fine mesh

	Experimental Result	CFD Result Time-Step Size 0.002	CFD Result Time-Step Size 0.0002	CFD Result Time-Step Size 0.00002
Frequency (Hz)	-	10.2	9.4	8.2
Error	-	-	-	-

Since in the simulations at which the smallest available time step size, meaning 0.00002, was employed, the associated frequencies of the pressure fluctuations were predicted with the least amount of uncertainty in comparison with the other two case studies, this time-step size was recognized as the best simulation parameter in the current research. Using all the selected numerical parameters and a monitor point, placed at $x/c = 0.4$ on suction side of the hydrofoil, the numerical results were extracted and the Power Spectral Density diagrams, showing the frequency of pressure fluctuations was obtained for four various cavitating flows (Figure 4.4). It is noteworthy that the magnitude is reported in y-axis in decibel (dB), by multiplying $\log_{10}(\text{magnitude})$ by factor 10, and the frequency is presented in the x-axis.

In the logarithmic magnitude diagram of f (Hz), $10 \log \left[\frac{S_{pp}(f)}{(\frac{1}{2}\rho U^2)^2} \right]$ (dB), showed in the Figure 4.4, the values of frequencies associated with the increase of pressure fluctuation intensity were predicted with very low error percentage. However, in the cavitating flow at $\sigma = 1.34$, the peak values of logarithmic magnitude were overpredicted in comparison with the experimental results.

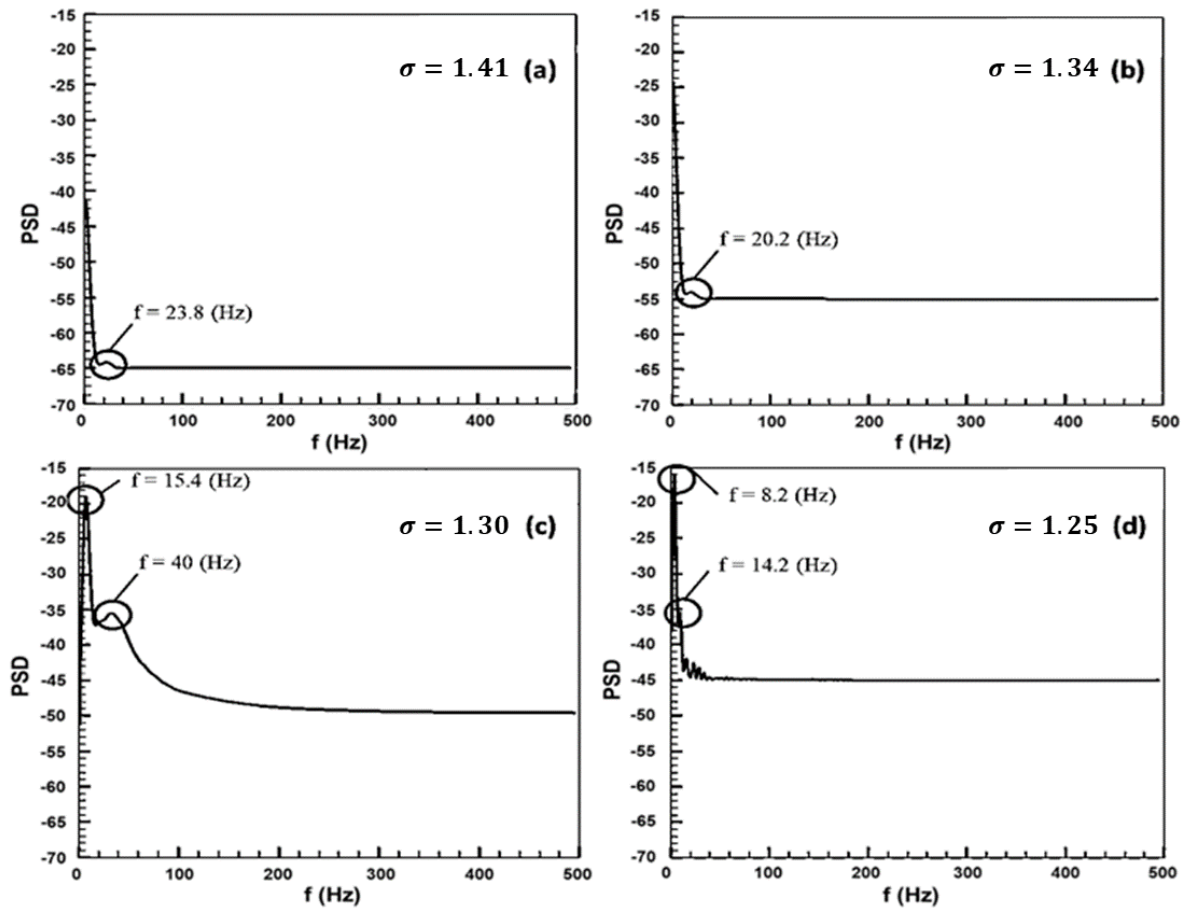


Figure 4.4 Power spectral density, obtained by performing transient analysis with time-step size of 0.00002 and monitor point placed at $x/c = 0.4$ in cavitating flow at various cavitation number

Figure 4.4 shows that as the cavitation number reduces, the frequency associated with pressure fluctuations reduces. It is also illustrated in Figure 4.4 that the pressure fluctuations increase as the cavitation intensity increases. For stronger cavitating flows, other dominant frequencies were detected. In the Figure 4.4(c), it is also shown that when the cavity length was $l/c = 0.5$, a frequency of 41 (Hz) is also predicted which is close to the experimentally measured value of 40 (Hz). The results in Figure 4.4 reveal that the predicted values are in good agreement with the experimental results.

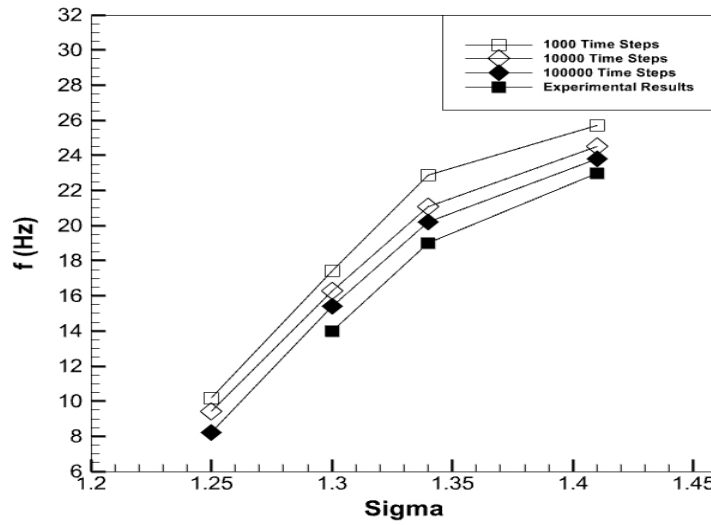


Figure 4.5 Numerical study on the effect of time-step size on the precision of the predicted frequencies associated to pressure fluctuations at $x/c = 0.4$ on the hydrofoil surface, in cavitating flows with various cavitation numbers

Based on their spectral analysis, Leroux et al. [9] indicated that the more cavity length increased, the more peak frequency of pressure oscillations decreased. Based on the data presented in the Figure 4.5, the current numerical predictions also confirmed this trend.

According to the Figure 4.5, the time step size has a significant effect on the precision of the predicted frequencies, associated to the pressure fluctuations, and the results obtained using a solution with the smallest available time-step size, namely 0.00002 (s) or 100000 time-steps, are in the best agreement with the experimental results.

CHAPTER 5 CONCLUSION

5.1 Conclusion and contribution

Numerical prediction of the cavitation phenomenon paved the way for manufacturers to avoid erosion of the blades of hydraulic machines and prevent subsequent costly maintenance. Numerical simulations can also save a large amount of time and money in some large-scale industrial projects in which it is costly and time-consuming to study cavitation experimentally.

In this research work, steady and unsteady numerical simulations coupled with a mixture multiphase approach were performed to find a reliable, robust, and more affordable alternative to predict the fluid characteristics in cavitating fluids with various cavitation numbers over a NACA66 hydrofoil. In this regard, ANSYS Fluent 16.1 was chosen as the solver and the effect of several parameters such as mesh density, turbulence model, cavitation model, and convergence criteria on the steady-state simulations, and time step size in transient simulation, was investigated for the accuracy of the numerical predictions in several case studies. It turned out that the standard $k - \varepsilon$ turbulence model and Kubota cavitation model were the most compatible models in the current numerical setup, and the numerical results were in very good agreement with the experimentally measured values when the convergence criteria were set to be $1e - 05$, and the time-step size was $2e - 05$. Then, the predicted static surface pressure coefficient (C_p) around the hydrofoil was compared with measurements by Leroux et al. [9]. Our numerical predictions demonstrated adequate agreement with the experimental results for most of the operating conditions studied.

Furthermore, the Power Spectral Density (PSD) diagrams of the sudden pressure distribution changes in the fluid flows at various cavitation numbers were obtained using FFT analysis. The predictions are quite close to the measured values. It should be noted that the error associated with the numerical predictions using a fine mesh with almost one million nodes, was varied between 3.5% and 10%, where the highest error was related to the case study in which the cavitating flow at $\sigma = 1.30$, meaning the most cavitant flow was modeled.

5.2 Limitations of the present work

However, in the current simulation using RANS turbulence model, even if the initiation location for cavitation, static pressure coefficient distribution around the hydrofoil, and the frequencies associated with the high-pressure fluctuations were all in good agreement with the experimental results, the cavity shedding phenomenon was not adequately predicted in the wake region after the trailing edge of the hydrofoil. This issue is related to the fact that the viscosity on the surface of hydrofoil is over estimated. In line with the comments made in the introduction chapter, it should be moderated by lowering the eddy-viscosity on the hydrofoil's surface in another study.

5.3 Recommendations for the future studies

Although, numerical results obtained using the standard $k - \varepsilon$ turbulence model demonstrated predictions with a decent precision, it is recommended that in further studies the effect of even denser mesh as well as turbulence models with a greater number of equations to be investigated. To obtain reliable predictions within the shortest feasible computation period, it is vitally important to find a good balance between the mesh density, time-step size, and turbulence model. In this regard, the LES method is highly recommended to be employed in comparison with the RANS and URANS methods to investigate the impact of turbulence on modeling the cavitation phenomenon. Furthermore, the effect of fluid compressibility should be calculated in another simulation on a supercomputer and using the density-based solver in the ANSYS-Fluent package. To implement new models without any limitation and to reduce the computational costs, it is recommended that the open source OpenFOAM solver be considered.

On the other hand, certain inaccuracies may be associated with the experimentally measured results and according to daily progress of measurement tools, experimental results must be recurring to verify that they represent the least measurement error in the behaviour of cavitating flow.

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APPENDIX A MESH QUALITY CRITERIA

Aspect ratio is the ratio of the longest side to the shortest sides in a computational mesh. However, the value of aspect ratio should be ideally close to 1:1, and it is recommended to keep this value below 100:1. However, in CFD it is not unusual to have an aspect ratio of more than 1000:1, and a mesh with such aspect ratio could potentially provides reliable predictions. As a rule of thumb, the computational grid should be as fine as it could capture all the gradients in the fluid flow. In the current research, using a mesh with relatively large aspect ratio value meaning 130, we ended up predicting accurate numerical results. However, decent predictions could be made using a mesh with larger values of aspect ratio, the calculations performed using a mesh with larger aspect ratio may have some limitations such as higher computation time, and sometimes interpolation errors.

The Orthogonal Quality is another mesh metrics to assess mesh quality. Orthogonality is defined as the angle ($^{\circ}$) between the connector vector of the cell-centers of two adjacent cells and the normal vector of the face, and it lies in the range of $[0,180]$, where the value of 0° corresponds to the best condition. Whereas orthogonality quality is defined as θ_{cell} and it should be calculated using the equation (58) in the ANSYS Fluent and CFX commercial solvers.

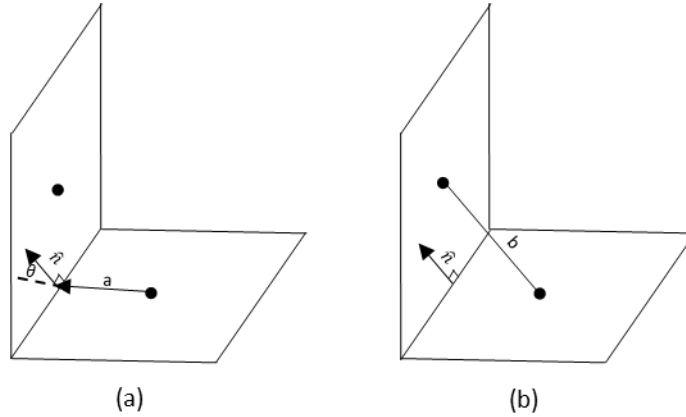


Figure 5.1 The possibilities to calculate θ_{cell} when the cells are not aligned with each other when the connecting vector is the vector defined between (a) the cell centroid and cell face, and (b) two cell centroids.

$$\theta_{cell} = \max \left[\cos^{-1} \left(\frac{a \cdot \hat{n}}{|a||\hat{n}|} \right), \cos^{-1} \left(\frac{b \cdot \hat{n}}{|b||\hat{n}|} \right) \right] \quad (58)$$

It should be noted that θ_{cell} is normalized to give the values between 0 and 1. It is recommended by ANSYS that the minimum orthogonal quality of 0.1 should be respected to have reasonable predictions. In the current work the minimum value of mesh orthogonal quality for the fine mesh is 0.38. According to the Table 5.1, this value represents a mesh with good quality in terms of orthogonality.

Table 5.1 Orthogonal Quality mesh metrics spectrum in ANSYS-Fluent commercial solver [59].



Unacceptable	Bad	Acceptable	Good	Very good	Excellent
0-0.001	0.001-0.14	0.15-0.20	0.20-0.69	0.70-0.95	0.95-1.00

The smoothness (σ) of a computational mesh is closely depended on the gradual variation of cell size, and sudden variations in size of the adjacent cells in a computational grid, make it less smooth and it may cause inaccurate results at the connecting nodes due to large truncation error. According to the recommendation of ANSYS for this mesh criterion, a smoothness in the range of [1.0,1.5] represents a good quality mesh, [1.5,2.5] represents a fair quality, and larger value of σ are representing poor mesh based on the definition of the smoothness mesh quality criterion.

Indeed, it could be a potential reason for uncertainty of the numerical results obtained using some hybrid meshes which they are not as precise as the results obtained using a structured mesh with is consisted of almost equal number of cells.

Finally, the skewness which is characterised by a number between 0 and 1, is another appropriate sign of the mesh quality, where a mesh with skewness of near 0 is the most ideal one, and as this value increases towards 1, the quality of mesh decreases. Generally, it is recommended by ANSYS to keep the skewness of mesh less than 0.95 so that the solver could predict reliable results. It is noteworthy that in the current study, maximum skewness value is 0.61 which also represents a mesh with good quality in terms of skewness Table 5.2.

Table 5.2 Skewness mesh metrics spectrum in the ANSYS Fluent commercial solver [59].

					
Excellent	Very good	Good	Acceptable	Bad	Unacceptable
0-0.25	0.25-0.50	0.50-0.80	0.80-0.94	0.95-0.97	0.98-1.00

To calculate the value of skewness in a computational mesh compounded of all cell types such as hexahedron, prism, and pyramid, using normalized angle deviation formula presented in equation (59), recognized as the best method, where, the values of θ_{max} and θ_{min} are illustrated in Figure 5.2, and θ_e is the angular face/cell, and its value is considered to be 90° for the quadrilateral and hexahedron, and 60° for the triangle and tetrahedron cells.

$$\text{Skewness} = \max\left[\frac{\theta_{max}-\theta_e}{180-\theta_e}, \frac{\theta_e-\theta_{min}}{\theta_e}\right] \quad (59)$$

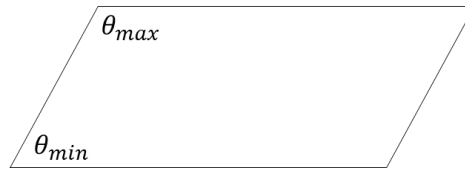


Figure 5.2 The minimum and maximum angle in a random cell to be used for calculating skewness of the computational mesh

APPENDIX B CONVERGENCE PLOTS

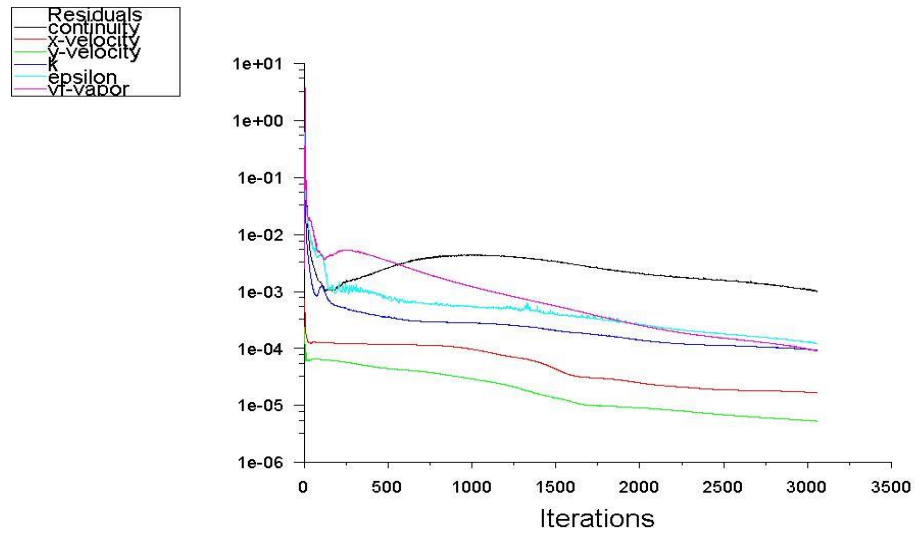


Figure 5.3 Scaled residual diagram related do the steady-state simulation of non-cavitating flow converged in around 3000 iterations

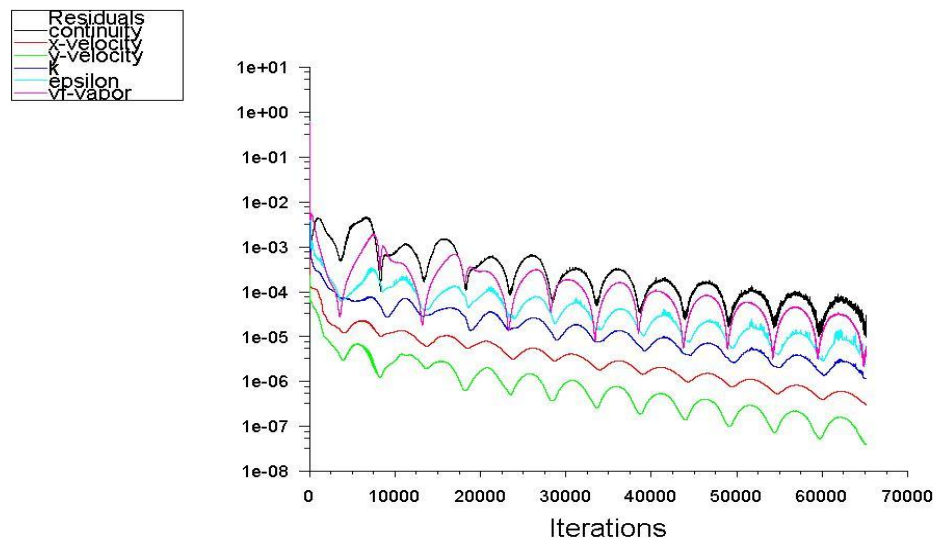


Figure 5.4 Scaled residual diagram related to the steady-state simulation of cavitating flow at $\sigma = 1.41$ converged in around 65000 iterations

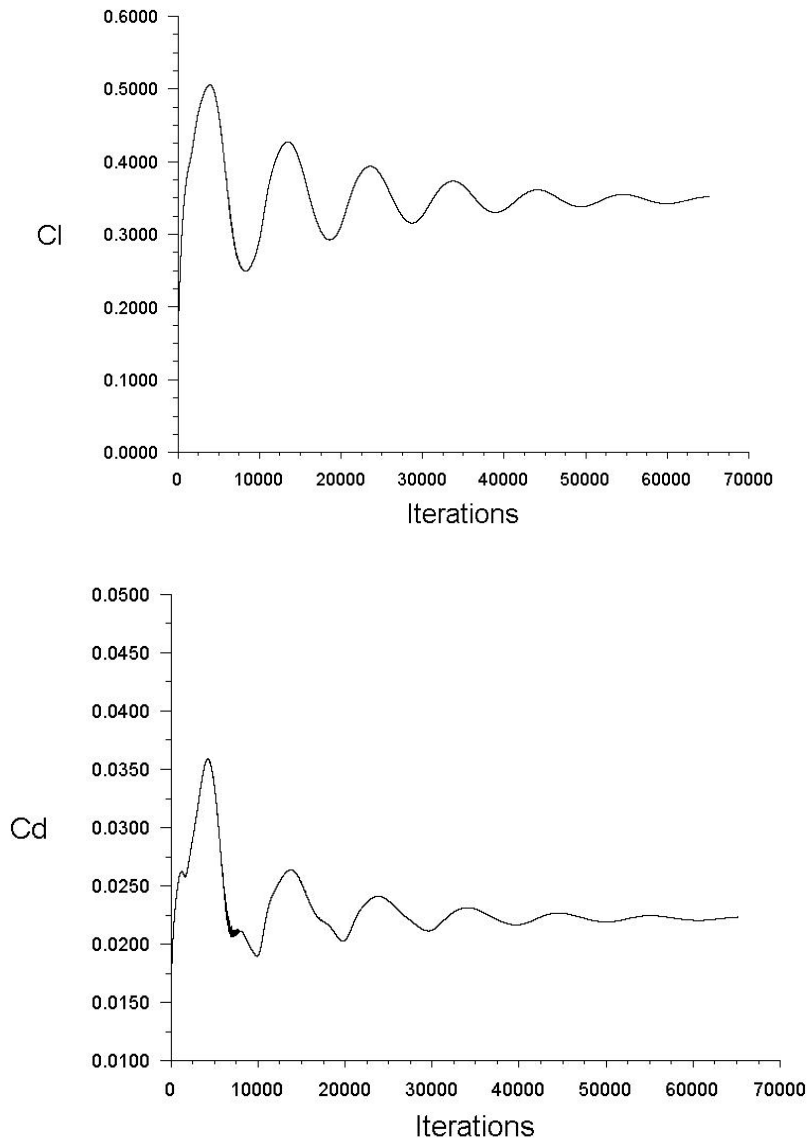


Figure 5.5 Diagrams of lift and drag coefficient related to the steady-state simulation of cavitating flow at $\sigma = 1.41$